PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients Undergoing Stent Implantation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials
AUTHORS	Shi, Yunmin; Wang, Yujie; Sun, Xuejing; Tang, Yan; Jiang, Mengqing; Bai, Yuanyuan; Liu, Suzhen; Jiang, Weihong; Yuan, Hong; Lu, Yao; Cai, Jingjing

VERSION 1 – REVIEW

REVIEWER	Henriques, Jose Academic Medical Center, Cardiology
REVIEW RETURNED	30-Sep-2020

GENERAL COMMENTS	This is one of so many meta-analyses on the topic. I fail to see the new insights.

REVIEWER	Melchior Seyfarth Witten/Herdecke University; Germany
REVIEW RETURNED	02-Oct-2020

GENERAL COMMENTS	To the authors: The paper „Effects of Percutaneous Mechanical Circulatory Support Devices in Acute Myocardial Infarction Patients in the Stenting Era: A Systematic Review and MetaAnalysis of Randomized Controlled Trials" by Cai et al focus on the impact of mechanical circulatory support in patients with acute myocardial infarction treated by PCI including coronary stents. This issue is under debate for a long time. Several small (n < 100) and mediumsized (n < 1000) RCT have been performed and multiple meta-analyses have been undertaken to elucidate the question whether mechanical support is of any use in patients with either acute myocardial infarction and/or cardiogenic shock. In this context, we have to put the present analysis by Cai et al. Although this issue has been addressed earlier in some meta-analyses the present systematic review and meta-analysis offers new insight since it sheds light on two issues: 1) All patients with ACS were included (with or without CS) 2) Only RCT including patients with PCI and stenting as the method of

	revascularization were examined. The first issue is of importance since some of the earlier meta-analyses focused only on patients with CS. The second issue is of major importance since only the contemporary treatment (state of the art) of patients with acute myocardial infarction is evaluated. This is important since IABP is much longer under discussion than revascularization with PCI and Stenting is the accepted treatment in acute myocardial infarction. So, it is reasonable to evaluate only RCT with stenting as the revascularization modality. Actually, there is no clear cut to describe a RCT as reflecting the state of the art treatment. However, to do that by defining stenting as inclusion criterium is a nice step and a good approximation to the contemporary treatment. Although these arguments support a publication of the meta-analysis of Cai et al, there are major issues to be addressed carefully before publication: 1. The conclusion that IABP reduces heart failure is neither supported by the data in the meta-analysis nor based on strong RCT. Heart Failure was never a primary endpoint in the evaluated trials. At the end, only 3 of 11 RCT have reported this secondary endpoint. Furthermore, the confidence interval for the impact of IABP on heart failure is 0.31 - 1.0 and therefore not supporting the conclusion. This conclusion can not be drawn and should be deleted. 2. There are too many inaccuracies in the paper. A systematic review is based on a profound, accurate and trustworthy work. The reviewer has found so many inaccuracies that a thorough revision seems to be necessary. Examples are given below: - Citations are wrong: In Table 1 the study SEMPER FI gets a 13. Actually, it is a 16; IABP-SHOCK gets a 7 (which is correct) and a 21 (which is incorrect). CRISP AMI gets a 16. However, 19 is correct. And so on …. - Prodzinsky et al performed their study in Germany not US - Seyfarth et al performed their study in Germany not US - IMPRESS was published in 2017 (not 2016) These are only some errors detected by the reviewer. A thorough revision is recommended. 3. The aim of the study to focus on the impact of circulatory support on top of contemporary treatment (stenting) in patients with AMI should be made more clear in the introduction. Otherwise, exclusion of RCT 13-16 is not understandable (see figure 1).

	4. On page 9, the term PCI and PTCA should be used more clearly. The authors mean PCI with stenting compared to PTCA (without stenting). 5. On page 13 (discussion), line 30 ff the meta-analysis of Thiele et al (Eur Heart J 2017) should be cited since it focus on the impact of pVAD in CS patients. 6. On page 13, the authors discuss the hypothesis that patients with left coronary artery related infarcts had a benefit by IABP. This is purely speculative and should be deleted or discussed more cautiously.

REVIEWER	Rita Pavasini UO Cardiologia Azienda Ospedaliero Universitaria di Ferrara Ferrara Italy
REVIEW RETURNED	07-Jan-2021

GENERAL COMMENTS	This is a meta-analysis about outcome of patients with myocardial infarction treated with IABP or other percutaneous mechanical support. I have some suggestions to optimize methods. - Please report also in abstract the I2 in every outcome presented. - Title of the meta-analysis is misleading: the meta-analysis is about the role of IABP in patients with acute myocardial infarction with or without circulatory support. I suggest to change the tile. - In European Guidelines the use of IABP is in class IIaC- IIIA so I would recommend to change the sentence about class of recommendation in introduction. - Please quote also: doi: 10.1097/CCM.0000000000002692 as background for this meta-analysis focused on RCT. - Which are the variables extracted by every study included? Please specify in methods. - Which kind of bias have been analyzes in quality assessment, please specify. - The degree of I2 is not correct. Please see: BMJ 2003;327:557–560. - It is not necessary to perform fixed vs random effect considering that the assumption to use the fixed effect is incorrect for studies with different study design (e.g: different mechanical circulatory support used), the leave one out approach has to be used. - Egger's intercept and Begg and Mazumdar's text should have been assessed for publication bias analysis. - Please cite in text where are summarized other patients population characteristics (e.g. table 1 and 2). - I2: 39% is a low heterogeneity (<25% is insignificant). - Please reported the number of patients included in each analysis, as well as the number of events per group. - Please report for each study the centre of the enrolment and the time of the enrolment (to verify any potential duplicate of the sample population). - In figure about search strategy please revise numbers, record screened by title and abstract should be 5207 if duplicate are 6670. - Figures (bottle funnel and forest plot) need caption and legend in supplemental online material, I would suggest to perform a word file with figure legend as supplemental online material.

REVIEWER	Antonio Nenna Università Campus Bio-Medico di Roma, Rome, Italy
REVIEW RETURNED	17-Jan-2021

GENERAL COMMENTS	Authors should be congratulated for their work.

VERSION 1 – AUTHOR RESPONSE

Point-by-Point Response to Reviewer 1's Comments

Question: This is one of so many meta-analyses on the topic. I fail to see the new insights.

Response: Thanks for the useful comment. To our knowledge, it is the first study that evaluate the efficacy and safety of mechanical circulatory support (MCS) in patients with AMI undergoing stent implantation. Unlike other meta-analyses focus on AMI patients, these meta-analyses also included patients with early systemic thrombolysis[1,2,3,4] and balloon angioplasty[3,5], this meta-analysis only included AMI patients with stenting. As rapid advances in percutaneous coronary intervention (PCI) technology, stenting is better for reperfusion after AMI than balloon angioplasty with lower acute risk of major complications and late-term restenosis [6,7]. So, it may not be suitable to continue to include stent implantation and balloon angioplasty as PCI in the studies of MCS, which the revascularization strategy may bring large heterogeneous. Therefore, it is reasonable and necessary to evaluate RCT with stenting as the revascularization modality. The new insights were summarized as the follows:

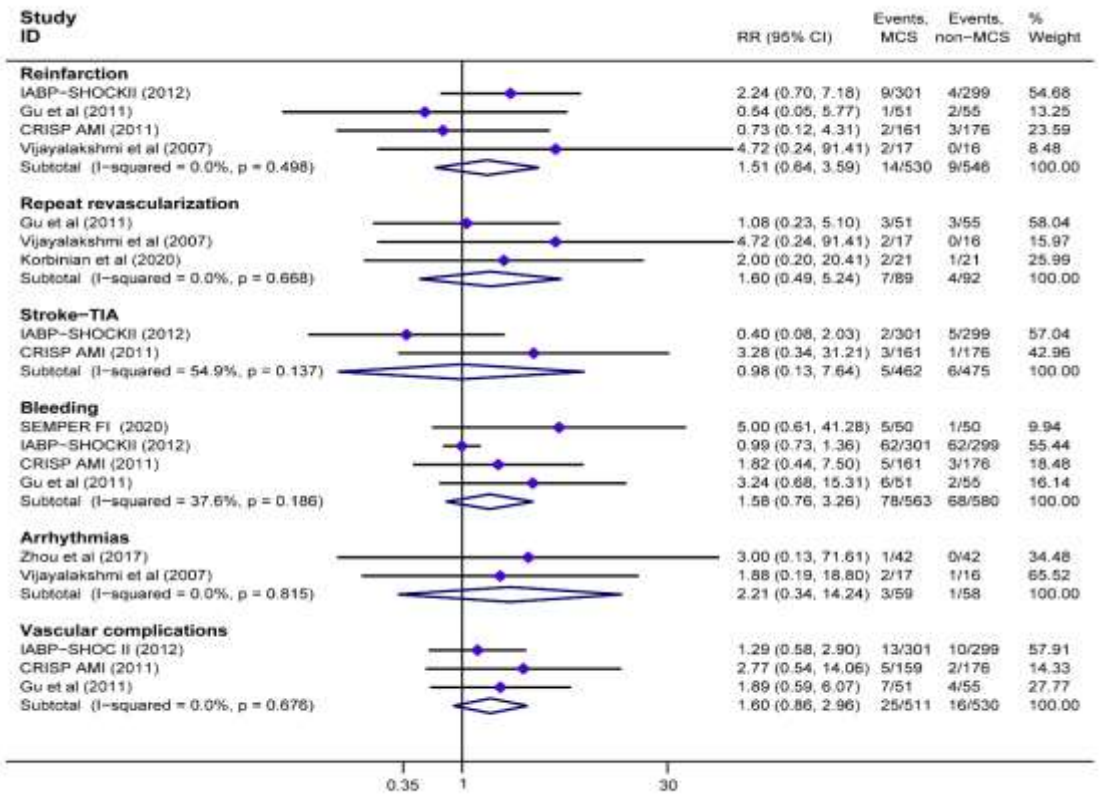
1. This meta-analysis focuses on the effect of MCS in AMI with stenting.
2. This meta-analysis generates a promising hypothesis that IABP reduced long-term (at least 6 months) mortality by half in the AMI patients with cardiogenic shock receiving stenting treatment (5.0% versus 10.3%). This benefit does not exist in the patients with CS. Studies with randomized control and decent sample size are warranted to confirm this finding.
3. These results have great implications for the choice of treatment in patients with AMI.

Point-by-Point Response to Reviewer 2's Comments

Question 1: The conclusion that IABP reduces heart failure is neither supported by the data in the meta-analysis nor based on strong RCT. Heart Failure was never a primary endpoint in the evaluated trials. At the end, only 3 of 11 RCT have reported this secondary endpoint. Furthermore, the confidence interval for the impact of IABP on heart failure is 0.31 - 1.0 and therefore not supporting the conclusion. This conclusion can not be drawn and should be deleted.

Response: Thanks for these helpful suggestions. We appreciate and agree with these suggestions. We have reviewed our paper and checked each study's definition of heart failure, found that the definition of heart failure previous paper was not completely correct. Actually, heart failure was not a

primary endpoint in these trials. Thus, we have deleted heart failure as an endpoint evaluated. In '**Endpoints**' of '**Methods**' on page 7, heart failure as a secondary endpoint and the definition have been deleted; in '**Abstract**' on page 2 to 3, the result and conclusion of heart failure have been deleted; in '**Outcomes of Synthesis Analyses**' of '**Results**' on page 10, the result of heart failure have been deleted; in '**Discussion**' and '**Conclusions**'on page 12 to 15, the conclusion of heart failure have been deleted as well. And we have also changed the Figure (**Online supplementary file 6**) as the follow :



Online supplementary file 6: Forest plots comparing the secondary end points between AMI patients treated with MCS and without MCS. TIA, transient ischaemic attack; MCS, mechanical circulatory support; non-MCS, no plan of MCS use during peri-operative period.

Question 2: There are too many inaccuracies in the paper. A systematic review is based on a profound, accurate and trustworthy work. The reviewer has found so many inaccuracies that a thorough revision seems to be necessary.

Response: Thanks for these helpful suggestions. We appreciate and agree with these suggestions. We also deeply believe that a systematic review must be based on a profound, accurate and trustworthy work. We have performed a systematic review profoundly, and found many inaccuracies in the paper, including citations, published year, region of studies, time point of reported mortality, follow-up, prior PCI and prior CABG. The revisions have been performed and show in table 1 and

table 2, and other tables have been revised (**Online supplementary file 3**, **Online supplementary file 5**, **Online supplementary file 8**, **Online supplementary file 9**). In consideration of multiple centers involving studies, we have deleted the 'region' column in **Table 1** and added a new table of the enrollment information to make sure no duplicate of the sample population (**Online supplementary file 4**). We have also made a systematic modify to the whole article, including wrong citations and inappropriate descriptions. The changes are shown as the follows:

Table 1. Study Design of the Included Studies

Study	Year	MCS group / non-MCS group	No. of Patients	Stenting, %	BMS, %	PCI success, %	Timing of MCS	Time point of reported mortality	Follow-Up
SEMPER FI[25]	2020	IABP+ stenting/stenting	50/50	100	NA	90/96	After Stenting	30 days, 6 months	6 months
Zhou et al[29]	2017	IABP+ stenting/stenting	42/42	100	NA	NA	Before Stenting	In hospital, 12 months	12 months
IABP-SHOCK II[8,9]	2012, 2013	IABP+ stenting/stenting	301/299	89.8	50.2/51.5	82.2/81.5	Before or after Stenting	30 days, 6 months	6 years
CRISP AMI[28]	2011	IABP+ stenting/stenting	161/176	96.9	53.7/53.1	92.9/95.3	Before Stenting	30 days, 6 months	6 months
Gu et al[30]	2011	IABP+ stenting/stenting	51/55	100	NA	NA	Before Stenting	30 days, 6 months	6 months
Prondzinsky et al[31]	2010	IABP+ stenting/stenting	19/21	85	NA	NA	After Stenting	In hospital	In hospital

Arias et al[33]	2005	IABP+ stenting/stenting	31/9	100	NA	NA	Before Stenting	In hospital	In hospital
Vijayalakshmi et al[34]	2007	IABP+ stenting/stenting	17/16	100	NA	NA	After Stenting	In hospital, 30 days	30 days
IMPRESS[26]	2017	pVADs+ stenting /IABP+ stenting	24/24	97.9	4/8	NA	After Stenting	30 days, 6 months	6 months
Seyfarth et al[32]	2008	pVADs+ stenting /IABP+ stenting	13/13	NA	NA	100/92	After Stenting	30 days	30 days
Thiele et al[27]	2005	pVADs+ stenting /IABP+ stenting	21/20	95.1	NA	80/74	Before or after Stenting	30 days	30 days
Korbinian et al[35]	2020	ECLS+ stenting / stenting	21/21	100	NA	NA	After stenting	30 days, 12 months	12 months

MCS, indicates mechanical circulatory support; IABP, intra-aortic balloon pump; PCI, percutaneous coronary intervention, including stenting and balloon angioplasty; pVADs, percutaneous ventricular assist devices; ECLS, extracorporeal life support; BMS, bare-metal stent; NA, not available. PCI success was defined as TIMI III flow.

Table 2. Baseline Clinical Characteristics of the Included Studies

Study (year)	Male, %	Mean age, year	AMI, %	CS, %	Infarct-related artery (LM/LAD/LCX/RCA), %	Hypertension, %	Diabetes mellitus, %	Previous stroke/TIA, %	Prior MI, %	Prior PCI, %	Prior CABG, %
SEMPER FI[25]	74/78	64/63	100	0	1.0/50.0/11.0/33.0	24/32	8/18	2/2	6/10	8/12	2/8

Zhou et al[29]	67/62	57.2/56.7	100	100	0/34.5/38.1/27.4	NA	NA	NA	NA	NA	NA
IABP-SHOCK II[8,9]	70.6/67.1	70/69	100	100	9/42.2/18.7/25.3	72/66.6	35.4/30.1	8/6.7	23.7/20.4	21.1/17.4	6.7/4.0
CRISP AMI[28]	82/81.8	56.1/57.7	100	0	0/97.3/0/0.3	24.2/34.1	16.8/20.5	0/0.6	0/0	1.9/1.1	NA
Gu et al[30]	56.9/65.5	67.4/66.6	100	0	3.8/47.2/11.3/37.7	68.6/60	35.5/34.5	NA	3.9/5.5	68.6/70.9	NA
Prondzinsky et al[31]	74/81	62.1/66.1	100	100	NA	42.1/47.6	52.6/47.6	NA	21.1/23.8	NA	NA
Arias et al[33]	NA	NA	100	100	0/60.0/22.5/8.3	NA	NA	NA	NA	NA	NA
Vijayalakshmi et al[34]	82/88	40/45	100	0	NA	25/38	18/25	NA	12/25	NA	NA
IMPRESS[26]	75/83	58/59	100	100	6.3/64.6/18.8/10.4	20/29	9/13	0/4	5/4	NA	NA
Seyfarth et al[32]	62/85	65/67	100	100	NA	54/69	39/23	NA	NA	NA	NA
Thiele et al[27]	76/75	63/65	100	100	NA	76/75	52/55	NA	62/45	NA	NA
Korbinian et al[35]	76.2/95.2	62/70	100	100	9.5/54.8/16.7/19.0	55/75	15/30	0/10	15/30	20/20	0/0

AMI indicates acute myocardial infarction; CS, cardiogenic shock; LM, left main; LAD, left anterior descending; LCX, left circumflex; RCA, right coronary artery; TIA, transient ischemic attack; MI, myocardial infarction; CABG, coronary artery bypass graft.

Online supplementary file 3: The Endpoint Definitions of the Included Studies

Study	Endpoint	Definition
SEMPER FI[25]	Reinfarction	NA.

	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding events according to the Global Use of Strategies to Open Coronary Arteries (GUSTO) trial.
	Arrhythmias	NA.
	Vascular complications	NA.
Zhou et al[29]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	Malignant arrhythmia (not detailed).
	Vascular complications	NA.
IABP-SHOCK II[8,9]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke identified by the presence of new neurologic symptoms in conjunction with signs of ischemia or bleeding on computed tomography.
	Bleeding	Severe or life-threatening bleeding and moderate bleeding as assessed according to the Global Use of Strategies to Open Occluded Coronary Arteries (GUSTO) criteria1.
	Arrhythmias	NA.
	Vascular complications	Peripheral ischemic vascular complications requiring surgical or interventional therapy.
CRISP AMI[28]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke (not detailed).
	Bleeding	Major bleeding events defined by theGlobal Use of Strategies To Open Coronary Arteries (GUSTO) trial and transfusions.
	Arrhythmias	NA.
	Vascular complications	Major vascular complications defined to include limb ischemia requiring intervention, distal embolization, major dissection, pseudoaneurysm or arteriovenousfistula, hematoma larger than 5 cm, and amputations.
Gu et al[30]	Reinfarction	Reinfarction (not detailed).

	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Vascular complications (not detailed).
Prondzinsky et al[31]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmia (not detailed).
	Vascular complications	NA.
Arias et al[33]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.
Vijayalakshmi et al[34]	Reinfarction	Reinfarction was defined as chest pain associated with ST-segment re-elevation and an increase in cardiac enzymes (3 times creatine kinase release).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmias including ventricular tachycardia, ventricular fibrillation a atrial fibrillation.
	Vascular complications	Vascular complications (not detailed).
IMPRESS[26]	Reinfarction	Reinfarction Defined by 3rd definition MI. Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference lim (URL) and with at least one of the following: 1. Symptoms of ischemia. 2. New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB). 3. Development of pathological Q waves in the ECG. 4. Imaging evidence of new loss of viable myocardium or new regional wall

		motion abnormality. 5. Identification of an intracoronary thrombus by angiography or autopsy.
	Repeat revascularization	Repeat PCI or CABG.
	Stroke-TIA	Any stroke confirmed by a neurologist and a concurring CT scan.
	Bleeding	1.Bleeding with associated serum hemoglobin level decrease of at least 5 g/dL (=3.1 mmol/L). The decrease in hemoglobin will be calculated as the last recorded Hb measurement preceding the onse of the bleeding, subtracted by the nadir Hb measurement (associate with the bleeding). 2. Bleeding necessitating a minimum of 2 packed cells of blood product transfusion (only blood transfusions that are explicitly related to the bleeding are taken into account). 3. The nee for surgery to control the bleeding.
	Arrhythmias	Placement of implantable cardioverter-defibrillator.
	Vascular complications	1.A major bleed arising at the arterial access site (for major bleeding definition see above) requiring extraction of the device. 2. A thrombotic occlusion of the femoral artery. 3. Limb ischemia requirin extraction of either of the study devices. 4. The need for vascular surgery to correct a vascular complication.
Seyfarth et al[32]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Limb ischemia (not detailed).
Thiele et al[27]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	Cerebrovascular accidents with neurological dysfunction.
	Bleeding	Major bleedings requiring transfusion of blood components.
	Arrhythmias	NA.
	Vascular complications	Lower extremity ischaemia requiring surgical or interventional action
Korbinian et al[35]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.

	Vascular complications	NA.

NA indicates not available.

Online supplementary file 4: The Enrollment Information of the Included Studies

Study	Center of Enrollment	Time of Enrollment
SEMPER FI[25]	Catharina Hospital, Eindhoven, Netherlands	August 2014 to December 2018
Zhou et al[29]	Department of Cardiology of Liaocheng People's Hospital, Liaocheng, China	January 2014 to January 2016
IABP-SHOCK II[8,9]	University of Leipzig-Heart Center, Leipzig, Germany Heart Center Bad Krozingen, Bad Krozingen, Germany Asklepios Klinik Langen, Langen, Germany German Heart Center Munich, Munich, Germany Heart Center-Segeberger Kliniken, Bad Segeberg, Germany SLK Kliniken Heilbronn, Heilbronn, Germany University of Greifswald, Greifswald, Germany Martin-Luther-University Halle-Wittenberg, Halle (Saale), Germany Klinikum Links der Weser, Bremen, Germany Heart Center Ludwigshafen, Ludwigshafen, Germany Zentralklinik Bad Berka, Bad Berka, Germany University Clinic of Saarland, Homburg/Saar, Germany Friedrich-Schiller University Jena, Jena, Germany University of Lübeck, Lübeck, Germany Klinik Hennigsdorf, Hennigsdorf, Germany Otto-von-Guericke-University Magdeburg, Magdeburg, Germany Sana Heart Center Cottbus, Cottbus, Germany Heart Center Dresden, Dresden, Germany University of Würzburg, Würzburg, Germany Hospital München-Neuperlach, Germany	June 2009 to March 2012

	Krankenhaus der Barmherzigen Brüder, Regensburg, Germany	
	Charité Berlin Campus Benjamin Franklin, Berlin, Germany	
	Vivantes Klinikum am Urban, Berlin, Germany	
	Carl-Thiem-Klinikum Cottbus, Cottbus, Germany	
	University Rostock, Germany	
	Kerckhoff Klinik Bad Nauheim, Bad Nauheim, Germany	
	Helios Klinikum Erfurt, Erfurt, Germany	
	Charité Berlin-Campus Mitte, Berlin, Germany	
	Ludwigs-Maximilian University Munich-Hospital Großhadern, Germany	
	Vivantes Auguste-Viktoria-Klinikum, Berlin, Germany	
	Klinikum Weiden, Weiden, Germany	
	Klinikum Fulda, Fulda, Germany	
	Katharinenhospital Unna, Unna, Germany	
	Philippusstift Essen, Essen, Germany	
	Carl-von-Basedow-Klinikum Merseburg, Merseburg, Germany	
	Klinikum Lippe-Detmold, Detmold, Germany	
	Schwarzwald-Baar Klinikum VillingenSchwenningen, Villingen-Schwenningen, Germany	
CRISP AMI[28]	Flinders Medical Centre, Bedford Park, Australia Virga Jesse Hospital, Hasselt, Belgium Leeds General Infirmary, Leeds, England Saint Thomas Hospital, London, England Universitatsklinikum Leipzig, Leipzig, Germany University of Freiburg, Freiburg, Germany Klinik III Fur Innere Medizin der Universitat zu Koln, Dresden, Germany CARE Hospital, Hyderabad, India Baroda Heart Institute and Research Centre, Vadodara, India	December 2008 to March 2011

	Sir Ganga Ram Hospital, New Delhi, India Gurunanak CARE Hospital, Hyderabad, India Medanta-The Medicity, Gurgaon, India Tagore Hospital and Heart Care Centre, Jalandhar, India Indraprastha Apollo Hospital, New Delhi, India Apollo Gleneagles Hospital, Kolkata, India Saint John's Medical College and Hospital, Bangalore, India Mater Misercordiae Dublin, Dublin, Ireland Catherina Hospital, Eindhoven, Netherlands Royal Infirmary of Edinburgh, Edinburgh, Scotland Duke University Medical Center, Durham, United States University of Texas Health Science Center, Houston, United States Temple University Hospital, Philadelphia, United States Tufts Medical Center, Boston, United States Newark Beth Israel Medical Center, Newark, United States University of Rochester Medical Center-Strong Memorial Hospital, Rochester, United States Lancaster General Hospital, Lancaster, United States University of Alabama Medical Center, Birmingham, United States Washington Hospital Center, Washington, United States Allegheny General Hospital, Pittsburgh, United States University of Southern California University Hospital, Los Angeles, United States	
Gu et al[30]	Min Hang Hospital, Shanghai, China	January 2005 to February 2009
Prondzinsky et al[31]	Department of Medicine III, Department of Biometrics and Statistics (SU), Halle-Wittenberg, Germany	March 2003 to June 2004
Arias et al[33]	Departamento de Medicina Interna, Fundación Clínica Médica Sur, México	February 2001 to February 2003

Vijayalakshmi et al[34]	The James Cook University Hospital, Middlesbrough, United Kingdom	NA
IMPRESS[26]	Academic Medical Center, Amsterdam, Netherlands Haukeland University Hospital, Bergen, Norway	May 2012 to September 2015
Seyfarth et al[32]	Deutsches Herzzentrum München, Munich, Germany Medizinische Klinik rechts der Isar, Munich, Germany	August 2004 to March 2006
Thiele et al[27]	University of Leipzig-Heart Centre, Leipzig, Germany	August 2000 to December 2003
Korbinian et al[35]	University Hospital Munich, Munich, Germany	November 2015 to March 2018

Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	
	In hospital	1.37 (0.50-3.78)
	30 days	0.70 (0.39-1.25)
	Different types of MCS use	
	IABP vs non-IABP	0.92 (0.57-1.47)
	ECLS vs non-ECLS	0.33 (0.01-1.47)
	Excluding high risk bias studies	
	Zhou et al[29], Arias et al[33] and Korbinian et al[35]	0.85 (0.51-1.45)
	Excluding each of the studies	
	SEMPER FI[25]	0.92 (0.57-1.48)
	Zhou et al[29]	0.80 (0.53-1.21)
	IABP-SHOCK II[8]	0.89 (0.45-1.74)
	CRISP AMI[28]	0.95 (0.59-1.55)
	Gu et al[30]	1.02 (0.66-1.58)
	Prondzinsky et al[31]	0.80 (0.48-1.34)

	Arias et al[33]	0.98 (0.58-1.66)
	Vijayalakshmi et al[34]	0.86 (0.55-1.34)
	Korbinian et al[35]	0.92 (0.57-1.47)
Long-term mortality		
	Inconsistent length of follow-up	
	6 months	0.72 (0.43-1.20)
	12 months	0.90 (0.35-2.31)
	Different types of MCS use	
	IABP vs non-IABP	0.87 (0.60-1.25)
	ECLS vs non-ECLS	0.50 (0.18-1.41)
	Excluding high risk bias studies	
	Zhou et al[29] and Korbinian et al[35]	0.72 (0.43-1.20)
	Excluding each of the studies	
	SEMPER FI[25]	0.82 (0.56-1.20)
	Zhou et al[29]	0.70 (0.45-1.09)
	IABP-SHOCK II[9]	0.68 (0.40-1.17)
	CRISP AMI[28]	0.89 (0.65-1.23)
	Gu et al[30]	0.92 (0.66-1.28)
	Korbinian et al[35]	0.87 (0.61-1.25)

CI indicates confidence interval.

Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

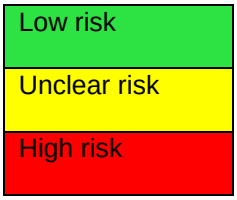
		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	
	IMPRESS[26]	0.97 (0.57-1.65)

	Seyfarth et al[32]	0.93 (0.59-1.46)
	Thiele et al[27]	0.94 (0.58-1.53)
Long-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	Not applicable

Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
SEMPER FI[25]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhou et al[29]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IABP-SHOCK II[8,9]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
CRISP AMI[28]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Gu et al[30]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Prondzinsky et al[31]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Arias et al[33]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Vijayalakshmi et al[34]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IMPRESS[26]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Seyfarth et al[32]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk

Thiele et al[27]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Korbinian et al[35]	Low risk	Low risk	Low risk	Low risk	Low risk	High risk	Low risk



Question 3: The aim of the study to focus on the impact of circulatory support on top of contemporary treatment (stenting) in patients with AMI should be made more clearly in the introduction. Otherwise, exclusion of RCT 13-16 is not understandable (see figure 1).

Response: Thanks for these helpful suggestions. We appreciate and agree with these suggestions. We have rewritten the introduction and highlighted this study focus on the impact of mechanical circulatory support on the contemporary treatment (stenting) in patients with AMI as follows:

As rapid advances in percutaneous coronary intervention (PCI) technology, stenting is better for reperfusion after AMI than balloon angioplasty [15]. However, it is noteworthy that both patient with stenting and balloon angioplasty were included in the studies of MCS [16,17]. As the revascularization strategy may bring large heterogeneous, the choice of the strategy must take into account the evidence. Therefore, in this study, we conduct a systematic review and meta-analysis to highlight the effect of MCS in AMI patients with stent implantation and further exploration on the effect of MCS are performed in the patients with or without CS. (in '**Introduction**' on page 5)

Question 4: On page 9, the term PCI and PTCA should be used more clearly. The authors mean PCI with stenting compared to PTCA (without stenting).

Response: Thanks for these useful suggestions. To avoid confusion, we have redefined PCI (including stenting and balloon angioplasty), replaced balloon angioplasty with the term of PTCA, revised the term of PCI with stent into stenting or stent implantation. The **Table 1** and **Figure 1** has been corrected as well. The changes are shown as the follows:

Studies in which patients underwent coronary artery bypass grafting (CABG), balloon angioplasty or systemic thrombolysis as the main revascularization method were also excluded. (in '**Study Selection**' of '**Methods**' on page 6)

Among these, 15 RCTs used PCI as the main revascularization therapy, 3 RCTs on balloon angioplasty use were excluded [22,23,24], and 12 RCTs finally met the inclusion criteria [8,25,26,27,28,9,29,30,31,32,33,34,35]. (in '**Results**' on page 8 to 9)

Some previous meta-analyses showed similar conclusions, but there was no detailed distinction between the revascularization strategies of thrombolytic therapy or PCI (balloon angioplasty and stent implantation) [16,17,41]. (in '**Efficacy of MCS in patients without CS**' of '**Discussion**' on page 13)

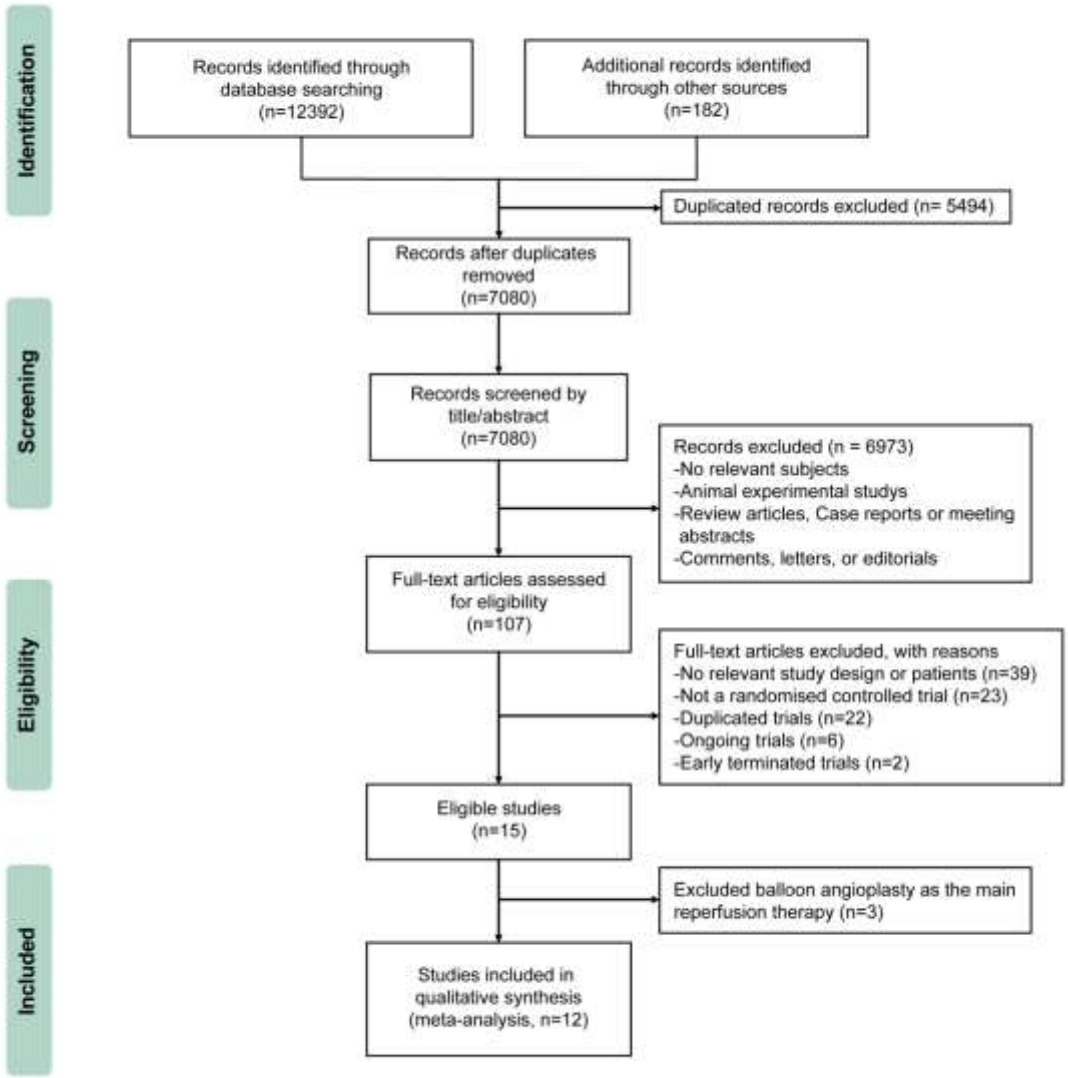


Figure 1. Flow diagram of study selection. Randomized clinical trials were included.

Table 1. Study Design of the Included Studies

Study	Year	MCS group / non-MCS group	No. of Patients	Stenting, %	BMS, %	PCI success, %	Timing of MCS	Time point of reported mortality	Follow-Up
SEMPER FI[25]	2020	IABP+ stenting/stenting	50/50	100	NA	90/96	After Stenting	30 days, 6 months	6 months
Zhou et al[29]	2017	IABP+ stenting/stenting	42/42	100	NA	NA	Before Stenting	In hospital, 12 months	12 months
IABP-SHOCK II[8,9]	2012, 2013	IABP+ stenting/stenting	301/299	89.8	50.2/51.5	82.2/81.5	Before or after Stenting	30 days, 6 months	6 years
CRISP AMI[28]	2011	IABP+ stenting/stenting	161/176	96.9	53.7/53.1	92.9/95.3	Before Stenting	30 days, 6 months	6 months
Gu et al[30]	2011	IABP+ stenting/stenting	51/55	100	NA	NA	Before Stenting	30 days, 6 months	6 months
Prondzinsky et al[31]	2010	IABP+ stenting/stenting	19/21	85	NA	NA	After Stenting	In hospital	In hospital
Arias et al[33]	2005	IABP+ stenting/stenting	31/9	100	NA	NA	Before Stenting	In hospital	In hospital
Vijayalakshmi et al[34]	2007	IABP+ stenting/stenting	17/16	100	NA	NA	After Stenting	In hospital, 30 days	30 days
IMPRESS[26]	2017	pVADs+ stenting	24/24	97.9	4/8	NA	After Stenting	30 days, 6	6 months

		/IABP+ stenting						months	
Seyfarth et al[32]	2008	pVADs+ stenting /IABP+ stenting	13/13	NA	NA	100/92	After Stenting	30 days	30 days
Thiele et al[27]	2005	pVADs+ stenting /IABP+ stenting	21/20	95.1	NA	80/74	Before or after Stenting	30 days	30 days
Korbinian et al[35]	2020	ECLS+ stenting / stenting	21/21	100	NA	NA	After stenting	30 days, 12 months	12 months

MCS, indicates mechanical circulatory support; IABP, intra-aortic balloon pump; PCI, percutaneous coronary intervention, including stenting and balloon angioplasty; pVADs, percutaneous ventricular assist devices; ECLS, extracorporeal life support; BMS, bare-metal stent; NA, not available. PCI success was defined as TIMI III flow.

Question 5: On page 13 (discussion), line 30 ff the meta-analysis of Thiele et al (Eur Heart J 2017) should be cited since it focus on the impact of pVAD in CS patients.

Response: Thanks for the useful suggestion. We have quoted this meta-analysis in '**Discussion**' on page 14 as the follow:

The results showed that pVADs do not reduce the all-cause mortality, but significantly increased the rate of bleeding. These results are consistent with previous which focus on the impact of pVADs in CS patients [6,13].

Question 6: On page 13, the authors discuss the hypothesis that patients with left coronary artery related infarcts had a benefit by IABP. This is purely speculative and should be deleted or discussed more cautiously.

Response: Thanks for these helpful suggestions. We appreciate and agree with these suggestions. Actually, the hypothesis that patients with left coronary artery related infarcts had a benefit by IABP was not rigorous. So, we have deleted it in '**Efficacy of MCS in patients without CS**' and '**Clinical implications**' of the '**Discussion**' on page 13 and page 15.

Point-by-Point Response to Reviewer 3's Comments

Question 1: Please report also in abstract the I^2 in every outcome presented.

Response: Thanks for the helpful suggestion. We have reported I^2 in every outcome presented in '**Results**' of '**Abstract**' on page 2 to 3 as follows:

Compared with non-MCS, MCS was not associated with short-term (within 30 days) (RR=0.90; 95% CI, 0.57-1.41; I^2=46.8%) and long-term (at least 6 months) (RR=0.82; 95% CI, 0.57-1.17; I^2=37.6%) mortality reductions. In the subset of patients without cardiogenic shock (CS), compared with non-MCS, the patients with IABP treatment significantly decrease the long-term mortality (RR=0.49; 95% CI, 0.27-0.90; I^2=0), but without the short-term mortality reductions (RR=0.51; 95% CI, 0.22-1.19; I^2=17.9%). While, in the patients with CS, the patients with MCS did not benefit the short-term (RR=1.09; 95% CI, 0.67-1.79; I^2=46.6%) or long-term (RR=1.00; 95% CI, 0.75-1.33; I^2=22.1%) survival. Moreover, the application of pVADs increased risk of bleeding (RR=1.86; 95% CI, 1.15-3.00; I^2=15.3%) compared to IABP treatment (RR=1.86; 95% CI, 1.15-3.00; I^2=15.3%).

And we have found and reported the I^2 in secondary endpoints and subgroup analyses in '**Outcomes of Synthesis Analyses**' and '**Outcomes of Subgroup Analyses**' of '**Results**' on page 10 to 11 as follows:

For the secondary endpoints, MCS was no effect the incidence of reinfarction (MCS=2.6%, 14/530; non-MCS=1.6%, 9/546; RR=1.51; 95% CI, 0.64-3.59; p=0.346; I^2=0, p=0.498), repeat revascularization (MCS=7.9%, 7/89; non-MCS=4.3%, 4/92; RR=1.60; 95% CI, 0.49-5.24; p=0.435; I^2=0, p=0.668), stroke-TIA (MCS=1.1%, 5/462; non-MCS=1.3%, 6/475; RR=1.51; 95% CI, 0.64-3.59; p=0.988; I^2=54.9%, p=0.137), bleeding (MCS=13.9%, 78/563; non-MCS=11.7%, 68/580; RR=1.58; 95% CI, 0.76-3.26; p=0.218; I^2=37.6%, p=0.186), arrhythmias (MCS=5.1%, 3/59; non-MCS=1.7%, 1/58; RR=2.21; 95% CI, 0.34-14.24; p=0.404; I^2=0, p=0.815), and vascular complications (MCS=4.9%, 25/511; non-MCS=3.0%, 16/530; RR=1.60; 95% CI, 0.86-2.96; p=0.134; I^2=0, p=0.676) (see online supplementary file 6). (on page 10)

Compared with IABP, pVADs increased the rate of bleeding (pVADs=66.7%, 30/45; IABP=36.4%, 16/44; RR=1.86; 95% CI, 1.15-3.00; p=0.011; I^2=15.3%, p=0.227) and trended towards increased vascular complications (pVADs=15.5%, 9/58; IABP=0, 0/57; RR=5.48; 95% CI, 0.96-31.11; p=0.055; I^2=0, p=0.671) (see online supplementary file 7). The results of the prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 8). (on page 11)

In the subset of patients without cardiogenic shock (CS), compared with the patients without MCS, the patients with MCS treatment significantly decrease the long-term (MCS=5.0%, non-MCS=10.3%; RR=0.49; 95% CI, 0.27-0.90; p=0.020; I^2=0, p=0.870) (Figure 3B), but without changed the outcome in the short-term (MCS=4.3%, non-MCS=8.1%; RR=0.51; 95% CI, 0.22-1.19; p=0.121; I^2=17.9%, p=0.301) (Figure 3A). In patients with CS, MCS did not reduce short-term mortality (MCS=35.7%,

non-MCS=35.2%; RR=1.09; 95% CI, 0.67-1.79; p=0.726; I^2=46.6%, p=0.100) (Figure 3A) or long-term mortality (MCS=45.6%, non-MCS=45.9%; RR=1.00; 95% CI, 0.75-1.33; p=0.984; I^2=22.1%, p=0.227) (Figure 3B). (on page 11)

Question 2: Title of the meta-analysis is misleading: the meta-analysis is about the role of IABP in patients with acute myocardial infarction with or without circulatory support. I suggest to change the tile.

Response: Thanks for the helpful suggestion. The aim of this meta-analysis focuses on the effect of mechanical circulatory support (MCS) in patients with AMI with stenting. We have updated the systematic search of the literature to February 2021 to find out if any new studies have been included. Finally, one RCT focus on extracorporeal life support (ECLS) in AMI patients with CS undergoing stent implantation has been included. In twelve RCTs included, there were three different types of MCS, including intra-aortic balloon pump (IABP), percutaneous ventricular assist devices (pVADs), and ECLS, eight of these comparing IABP with non-IABP, one study comparing ECLS with non-ECLS, three studies comparing pVADs with IABP. Although there are few RCTs on the use of different types of MCS in AMI patients, IABP is the main one. But we believe this meta-analysis is just the beginning. We are paying close attention to some ongoing clinical trials exploring the efficacy of using different types of MCS in AMI patients, including ECMO, pVADs or ECLS (NCT04184635, NCT03637205, NCT01633502, NCT03947619).

Question 3: In European Guidelines the use of IABP is in class IIaC- IIIA so I would recommend to change the sentence about class of recommendation in introduction.

Response: Thanks for the useful suggestion. We appreciate and agree with this suggestion. We have refered to the American and European guides carefully and changed the sentence about class of recommendation in '**Introduction**' on page 5 as the follow:

Therefore, in AMI patients with cardiogenic shock (CS), IABP and pVADs were recommended to class IIa and class IIb in US guidelines respectively, and IABP to class IIaC-IIIA in European guidelines [10,11].

Question 4: Please quote also: doi: 10.1097/CCM.0000000000002692 as background for this meta-analysis focused on RCT.

Response: Thanks for the helpful suggestion. We have quoted this meta-analysis in '**Introduction**' on page 5 as the follow:

Recent meta-analyses have focused on the use of MCS in patients with CS, but no benefits for survival was found [12,13,14].

Question 5: Which are the variables extracted by every study included? Please specify in methods.

Response: Thanks for the useful suggestion. We are very sorry for missing such critical information. We have added the variables extracted in '**Data Extraction and Quality Assessment**' of '**Methods**' on page 7 as the follow:

All extracted data from the included studies were independently collected in duplicate using a standardized data extraction form, which contained authors or trails' name, year of publication, study design (interventions, sample size, percentage of stenting and bare-metal stent, PCI success, timing of MCS, time point of reported mortality, follow-up), baseline characteristics of study population (sex, age, AMI, CS, infarct-related artery, hypertension, diabetes mellitus, previous stroke/TIA and MI, prior PCI and CABG), the endpoint definitions, the centre and the time of the study enrolment.

Question 6: Which kind of bias have been analyzed in quality assessment, please specify.

Response: Thanks for the suggestion. We appreciate and agree with this suggestion. The kinds of bias analyzed in quality assessment were specified in '**Data Extraction and Quality Assessment**' of '**Methods**' on page 7 to 8 as the follow:

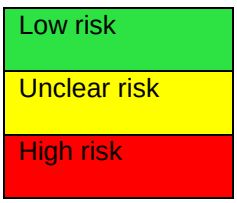
The quality of the included studies was assessed using the Cochrane Collaborations tool [19], which including selection bias, performance bias, detection bias, attrition bias, reporting bias, other possible bias. Each source of bias was classified as low, unclear, and high risk.

And we have also added the kinds of bias to the header of the column of the table '**Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials**' in **Online supplementary file 9:**

Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias

SEMPER FI[25]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhou et al[29]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IABP-SHOCK II[8,9]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
CRISP AMI[28]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Gu et al[30]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Prondzinsky et al[31]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Arias et al[33]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Vijayalakshmi et al[34]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IMPRESS[26]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Seyfarth et al[32]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Thiele et al[27]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Korbinian et al[35]	Low risk	Low risk	Low risk	Low risk	Low risk	High risk	Low risk



Question 7: The degree of I^2 is not correct. Please see: BMJ 2003;327:557–560.

Response: Thanks for the useful suggestion. We strongly agree with this suggestion. We have read this paper classifying the degree of heterogeneity by I^2 value carefully, which is very helpful to us. The degree of I^2 has been corrected in '**Statistical Analysis**' of '**Methods**' on page 8 as the follow:

Heterogeneity was categorized by I^2 as follows [20]: insignificant heterogeneity, I^2<25%; low heterogeneity, 25%≤I^2<50%; moderate heterogeneity, 50%≤I^2<75%; high heterogeneity, I^2≥75%.

And we have revised the descriptions heterogeneity in '**Outcomes of Synthesis Analyses**' of '**Results**' on page 10 as follows:

MCS was not associated with a short-term mortality reduction compared to the non-MCS (MCS=23.1%, 160/693; non-MCS=23.5%, 162/689; RR=0.90; 95% CI, 0.57-1.41; p=0.640), with low heterogeneity among studies (I^2=46.8%; p=0.059) (Figure 2). Similarly, there was no significant difference between the MCS group and the non-MCS group in the long-term mortality (MCS=28.6%, 176/626; non-MCS=30.3%, 195/643; RR=0.82; 95% CI, 0.57-1.17; p=0.276), with low heterogeneity among studies (I^2=37.6%; p=0.156) (Figure 2).

Question 8: It is not necessary to perform fixed vs random effect considering that the assumption to use the fixed effect is incorrect for studies with different study design (e.g: different mechanical circulatory support used), the leave one out approach has to be used.

Response: Thanks for the suggestion. We appreciate and agree with this suggestion. The sensitivity analyses by fixed versus random effects models have been deleted, and added two methods by inconsistent length of follow-up and different types of MCS use. The results of the prespecified sensitivity analyses have also been not significantly different, and been summarized in online **Online supplementary file 5** and **Online supplementary file 8**. The changes are shown as follows:

Sensitivity analyses were performed for primary outcomes with the following: (1) inconsistent length of follow-up, (2) different types of MCS use, (3) exclusion of research with a high risk of bias, and (4) the one-study-out method. (in '**Statistical Analysis**' of '**Methods**' on page 8)

Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	
	In hospital	1.37 (0.50-3.78)
	30 days	0.70 (0.39-1.25)
	Different types of MCS use	
	IABP vs non-IABP	0.92 (0.57-1.47)
	ECLS vs non-ECLS	0.33 (0.01-1.47)

	Excluding high risk bias studies	
	Zhou et al[29], Arias et al[33] and Korbinian et al[35]	0.85 (0.51-1.45)
	Excluding each of the studies	
	SEMPER FI[25]	0.92 (0.57-1.48)
	Zhou et al[29]	0.80 (0.53-1.21)
	IABP-SHOCK II[8]	0.89 (0.45-1.74)
	CRISP AMI[28]	0.95 (0.59-1.55)
	Gu et al[30]	1.02 (0.66-1.58)
	Prondzinsky et al[31]	0.80 (0.48-1.34)
	Arias et al[33]	0.98 (0.58-1.66)
	Vijayalakshmi et al[34]	0.86 (0.55-1.34)
	Korbinian et al[35]	0.92 (0.57-1.47)
Long-term mortality		
	Inconsistent length of follow-up	
	6 months	0.72 (0.43-1.20)
	12 months	0.90 (0.35-2.31)
	Different types of MCS use	
	IABP vs non-IABP	0.87 (0.60-1.25)
	ECLS vs non-ECLS	0.50 (0.18-1.41)
	Excluding high risk bias studies	
	Zhou et al[29] and Korbinian et al[35]	0.72 (0.43-1.20)
	Excluding each of the studies	
	SEMPER FI[25]	0.82 (0.56-1.20)
	Zhou et al[29]	0.70 (0.45-1.09)
	IABP-SHOCK II[9]	0.68 (0.40-1.17)
	CRISP AMI[28]	0.89 (0.65-1.23)
	Gu et al[30]	0.92 (0.66-1.28)
	Korbinian et al[35]	0.87 (0.61-1.25)

CI indicates confidence interval.

Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	
	IMPRESS[26]	0.97 (0.57-1.65)
	Seyfarth et al[32]	0.93 (0.59-1.46)
	Thiele et al[27]	0.94 (0.58-1.53)
Long-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	Not applicable

Question 9: Egger's intercept and Begg and Mazumdar's text should have been assessed for publication bias analysis.

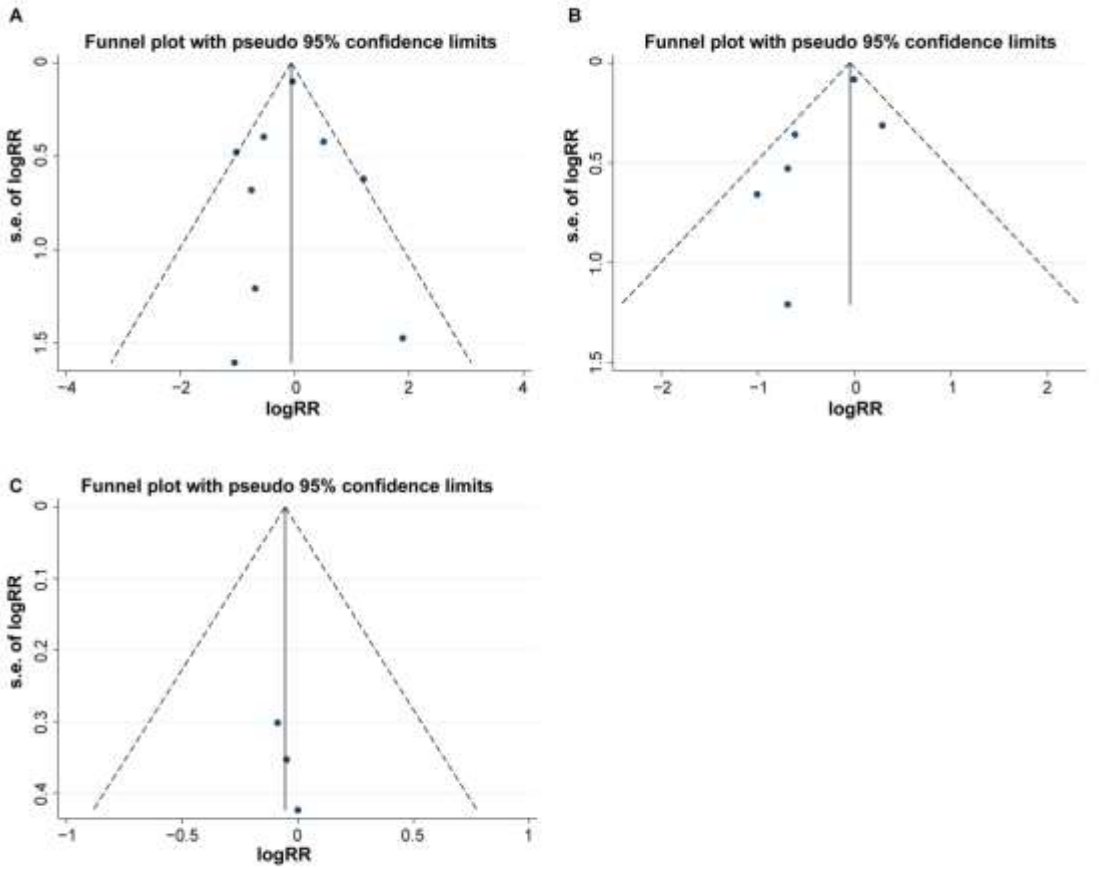
Response: Thanks for the useful suggestion. We strongly agree with this suggestion. We have assessed the publication bias analysis by Egger's intercept and Begg and Mazumdar's text, and found the publication bias by the Egger's test for short-term mortality of studies comparing pVADs use versus IABP use. So, we have refered to your previous meta-analysis and employed the trim-and-fill method to estimate the number of missing studies and RR adjustment. The descriptions have been revised as follows:

Funnel plot, Begg and Mazumdar rank correlation and Egger's regression intercept were used to evaluate publication bias [20]. If publication bias existed, the trim-and-fill method was performed for estimating the number of missing studies and RR adjustment [21]. (in '**Statistical Analysis**' of '**Methods**' on page 8).

There was no evidence of publication bias for included studies except the Egger's test for short-term mortality of studies comparing pVADs use versus IABP use (P<0.05) (online supplementary file 10

and online supplementary file 11). Then the trim-and-fill method was performed for estimation of the number of missing studies (pVADs vs IABP) that might exist. Result suggested 2 theoretically missing studies with an adjusted RR of short-term mortality (RR=0.92; 95% CI, 0.67-1.26; p=0.59) (in '**Study Quality and Publication Bias**' of '**Results**' on page 12).

Funnel plot in **Online supplementary file 10** and the results of Egger's intercept and Begg and Mazumdar's text summarized in **Online supplementary file 11** have been revised as follows:



Online supplementary file 10: Funnel plot of standard error for: A. MCS compared with non-MCS in short-term mortality; B. MCS compared with non-MCS in long-term mortality; C. pVADs compared with IABP in short-term mortality.

online supplementary file 11:

Publication bias on the association of primary outcome.

	Egger's test (P)	Begg's test (P)

MCS vs non-MCS			
	Short-term mortality	0.886	0.917
	Long-term mortality	0.154	0.452
pVAD vs IABP			
	Short-term mortality	0.015*	0.296
	Long-term mortality	Not applicable	Not applicable

* Have statistical difference.

Question 10: Please cite in text where are summarized other patients' population characteristics (e.g. table 1 and 2).

Response: Thanks for the useful suggestion. We have cited summarized patients' population characteristics in '**Characteristics of the Studies**' of '**Results**' on page 9 as the follows:

Among the 12 RCTs, 1497 AMI patients were included, 94.5% of patients (1391/1471) underwent stent implantation, 86.9% of patients (959/1104) were with PCI success. The follow-up period ranged from in-hospital only to 6 years.

The mean age of the population was 63.8, and 73.3% were male, 77.1% (1046/1357) had left coronary artery-related infarction, 52.9% (726/1373) had hypertension, 27.7% (381/1373) were diabetes mellitus, 4.3% (49/1172) had a previous diagnosis of stroke/TIA, 14.3% (193/1347) had a previous diagnosis of MI, 17.6% (208/1185) underwent PCI previously and 5.0% (37/742) underwent CABG previously.

Question 11: I^2: 39% is a low heterogeneity (<25% is insignificant).

Response: Thanks for the useful suggestion. We have revised the descriptions heterogeneity in '**Outcomes of Synthesis Analyses**' of '**Results**' on page 10 as follows:

MCS was not associated with a short-term mortality reduction compared to the non-MCS (MCS=23.1%, 160/693; non-MCS=23.5%, 162/689; RR=0.90; 95% CI, 0.57-1.41; p=0.640), with low heterogeneity among studies (I^2=46.8%; p=0.059) (Figure 2). Similarly, there was no significant difference between the MCS group and the non-MCS group in the long-term mortality (MCS=28.6%, 176/626; non-MCS=30.3%, 195/643; RR=0.82; 95% CI, 0.57-1.17; p=0.276), with low heterogeneity among studies (I^2=37.6%; p=0.156) (Figure 2).

Question 12: Please reported the number of patients included in each analysis, as well as the number of events per group.

Response: Thanks for the useful suggestion. We have reported the number of patients included and the number of events per group in each analysis in '**Characteristics of the Studies**', '**Outcomes of Synthesis Analyses**' and '**Outcomes of Subgroup Analyses**' of '**Results**' on page 9 to 11 as the follows:

Nine studies including 1382 AMI patients compared MCS with non-MCS (693 in the MCS group and 690 in the non-MCS group) [8,25,28,9,29,30,31,33,34,35], 3 studies including 115 AMI patients compared pVADs versus IABP (58 in the pVADs group and 57 in the IABP group) [26,27,32]. Eight RCTs included CS patients, among which 5 studies that included 806 patients compared MCS with non-MCS (414 in the MCS group and 392 in the non-MCS group) [8,9,29,31,33,35], the remaining 3 studies compared pVADs versus IABP. Four RCTs included patients without CS, and all types of MCS use were IABP, which included 576 patients, compared MCS with non-MCS (279 in the IABP group and 297 in the non-IABP group) [25,28,30,34]. (in'**Characteristics of the Studies**' of '**Results**' on page 9)

MCS was not associated with a short-term mortality reduction compared to the non-MCS (MCS=23.1%, 160/693; non-MCS=23.5%, 162/689; RR=0.90; 95% CI, 0.57-1.41; p=0.640), with low heterogeneity among studies (I^2=46.8%; p=0.059) (Figure 2). Similarly, there was no significant difference between the MCS group and the non-MCS group in the long-term mortality (MCS=28.6%, 176/626; non-MCS=30.3%, 195/643; RR=0.82; 95% CI, 0.57-1.17; p=0.276), with low heterogeneity among studies (I^2=37.6%; p=0.156) (Figure 2). The results of prespecified sensitivity analyses were not significantly different from those of the primary analysis (see online supplementary file 5). (in '**Outcomes of Synthesis Analyses**' of '**Results**' on page 10)

For the secondary endpoints, MCS was no effect the incidence of reinfarction (MCS=2.6%, 14/530; non-MCS=1.6%, 9/546; RR=1.51; 95% CI, 0.64-3.59; p=0.346; I^2=0, p=0.498), repeat revascularization (MCS=7.9%, 7/89; non-MCS=4.3%, 4/92; RR=1.60; 95% CI, 0.49-5.24; p=0.435; I^2=0, p=0.668), stroke-TIA (MCS=1.1%, 5/462; non-MCS=1.3%, 6/475; RR=1.51; 95% CI, 0.64-3.59; p=0.988; I^2=54.9%, p=0.137), bleeding (MCS=13.9%, 78/563; non-MCS=11.7%, 68/580; RR=1.58; 95% CI, 0.76-3.26; p=0.218; I^2=37.6%, p=0.186), arrhythmias (MCS=5.1%, 3/59; non-MCS=1.7%, 1/58; RR=2.21; 95% CI, 0.34-14.24; p=0.404; I^2=0, p=0.815), and vascular complications (MCS=4.9%, 25/511; non-MCS=3.0%, 16/530; RR=1.60; 95% CI, 0.86-2.96; p=0.134; I^2=0, p=0.676) (see online supplementary file 6). (in '**Outcomes of Synthesis Analyses**' of '**Results**' on page 10)

The pooled analysis revealed that pVADs was not associated with a decrease in short-term mortality compared with IABP (pVADs=44.8%, 26/58; IABP=47.4%, 27/57; RR=0.95; 95% CI, 0.64-1.41; p=0.786), with insignificant heterogeneity among studies (I^2=0; p=0.986) (see online supplementary file 7). Long-term mortality was not analysed because only one study reported. For the secondary outcome analysis, only bleeding and vascular complications met the quantitative requirements for the analysis. Compared with IABP, pVADs increased the rate of bleeding

(pVADs=66.7%, 30/45; IABP=36.4%, 16/44; RR=1.86; 95% CI, 1.15-3.00; p=0.011; I^2=15.3%, p=0.227) and trended towards increased vascular complications (pVADs=15.5%, 9/58; IABP=0, 0/57; RR=5.48; 95% CI, 0.96-31.11; p=0.055; I^2=0, p=0.671) (see online supplementary file 7). (in '**Outcomes of Synthesis Analyses**' of '**Results**' on page 10 to 11)

In the subset of patients without cardiogenic shock (CS), compared with the patients without MCS, the patients with MCS treatment significantly decrease the long-term (MCS=5.0%, non-MCS=10.3%; RR=0.49; 95% CI, 0.27-0.90; p=0.020; I^2=0, p=0.870) (Figure 3B), but without changed the outcome in the short-term (MCS=4.3%, non-MCS=8.1%; RR=0.51; 95% CI, 0.22-1.19; p=0.121; I^2=17.9%, p=0.301) (Figure 3A). In patients with CS, MCS did not reduce short-term mortality (MCS=35.7%, non-MCS=35.2%; RR=1.09; 95% CI, 0.67-1.79; p=0.726; I^2=46.6%, p=0.100) (Figure 3A) or long-term mortality (MCS=45.6%, non-MCS=45.9%; RR=1.00; 95% CI, 0.75-1.33; p=0.984; I^2=22.1%, p=0.227) (Figure 3B). (in '**Outcomes of Subgroup Analyses**' of '**Results**' on page 11)

Question 13: Please report for each study the centre of the enrolment and the time of the enrolment (to verify any potential duplicate of the sample population).

Response: Thanks for the useful suggestion. We appreciate and agree with this suggestion. We have systematically searched the enrolment information of each study, and summarized in online supplementary file 4. The center and the time of enrolment in each study were no duplicate of the sample population of included studies as the follow:

Online supplementary file 4: The Enrollment Information of the Included Studies

Study	Center of Enrollment	Time of Enrollment
SEMPER FI[25]	Catharina Hospital, Eindhoven, Netherlands	August 2014 to December 2018
Zhou et al[29]	Department of Cardiology of Liaocheng People's Hospital, Liaocheng, China	January 2014 to January 2016
IABP-SHOCK II[8,9]	University of Leipzig-Heart Center, Leipzig, Germany Heart Center Bad Krozingen, Bad Krozingen, Germany Asklepios Klinik Langen, Langen, Germany German Heart Center Munich, Munich, Germany Heart Center-Segeberger Kliniken, Bad Segeberg, Germany SLK Kliniken Heilbronn, Heilbronn, Germany University of Greifswald, Greifswald, Germany	June 2009 to March 2012

Martin-Luther-University Halle-Wittenberg, Halle (Saale), Germany

Klinikum Links der Weser, Bremen, Germany

Heart Center Ludwigshafen, Ludwigshafen, Germany

Zentralklinik Bad Berka, Bad Berka, Germany

University Clinic of Saarland, Homburg/Saar, Germany

Friedrich-Schiller University Jena, Jena, Germany

University of Lübeck, Lübeck, Germany

Klinik Hennigsdorf, Hennigsdorf, Germany

Otto-von-Guericke-University Magdeburg, Magdeburg, Germany

Sana Heart Center Cottbus, Cottbus, Germany

Heart Center Dresden, Dresden, Germany

University of Würzburg, Würzburg, Germany

Hospital München-Neuperlach, Germany

Krankenhaus der Barmherzigen Brüder, Regensburg, Germany

Charité Berlin Campus Benjamin Franklin, Berlin, Germany

Vivantes Klinikum am Urban, Berlin, Germany

Carl-Thiem-Klinikum Cottbus, Cottbus, Germany

University Rostock, Germany

Kerckhoff Klinik Bad Nauheim, Bad Nauheim, Germany

Helios Klinikum Erfurt, Erfurt, Germany

Charité Berlin-Campus Mitte, Berlin, Germany

Ludwigs-Maximilian University Munich-Hospital Großhadern, Germany

Vivantes Auguste-Viktoria-Klinikum, Berlin, Germany

Klinikum Weiden, Weiden, Germany

Klinikum Fulda, Fulda, Germany

Katharinenhospital Unna, Unna, Germany

Philippusstift Essen, Essen, Germany

	Carl-von-Basedow-Klinikum Merseburg, Merseburg, Germany	
	Klinikum Lippe-Detmold, Detmold, Germany	
	Schwarzwald-Baar Klinikum VillingenSchwenningen, Villingen-Schwenningen, Germany	
CRISP AMI[28]	Flinders Medical Centre, Bedford Park, Australia	December 2008 to March 2011
	Virga Jesse Hospital, Hasselt, Belgium	
	Leeds General Infirmary, Leeds, England	
	Saint Thomas Hospital, London, England	
	Universitatsklinikum Leipzig, Leipzig, Germany	
	University of Freiburg, Freiburg, Germany	
	Klinik III Fur Innere Medizin der Universitat zu Koln, Dresden, Germany	
	CARE Hospital, Hyderabad, India	
	Baroda Heart Institute and Research Centre, Vadodara, India	
	Sir Ganga Ram Hospital, New Delhi, India	
	Gurunanak CARE Hospital, Hyderabad, India	
	Medanta-The Medicity, Gurgaon, India	
	Tagore Hospital and Heart Care Centre, Jalandhar, India	
	Indraprastha Apollo Hospital, New Delhi, India	
	Apollo Gleneagles Hospital, Kolkata, India	
	Saint John's Medical College and Hospital, Bangalore, India	
	Mater Misercordiae Dublin, Dublin, Ireland	
	Catherina Hospital, Eindhoven, Netherlands	
	Royal Infirmary of Edinburgh, Edinburgh, Scotland	
	Duke University Medical Center, Durham, United States	
	University of Texas Health Science Center, Houston, United States	
	Temple University Hospital, Philadelphia, United States	
	Tufts Medical Center, Boston, United States	
	Newark Beth Israel Medical Center, Newark, United States	

	University of Rochester Medical Center-Strong Memorial Hospital, Rochester, United States Lancaster General Hospital, Lancaster, United States University of Alabama Medical Center, Birmingham, United States Washington Hospital Center, Washington, United States Allegheny General Hospital, Pittsburgh, United States University of Southern California University Hospital, Los Angeles, United States	
Gu et al[30]	Min Hang Hospital, Shanghai, China	January 2005 to February 2009
Prondzinsky et al[31]	Department of Medicine III, Department of Biometrics and Statistics (SU), Halle-Wittenberg, Germany	March 2003 to June 2004
Arias et al[33]	Departamento de Medicina Interna, Fundación Clínica Médica Sur, México	February 2001 to February 2003
Vijayalakshmi et al[34]	The James Cook University Hospital, Middlesbrough, United Kingdom	NA
IMPRESS[26]	Academic Medical Center, Amsterdam, Netherlands Haukeland University Hospital, Bergen, Norway	May 2012 to September 2015
Seyfarth et al[32]	Deutsches Herzzentrum München, Munich, Germany Medizinische Klinik rechts der Isar, Munich, Germany	August 2004 to March 2006
Thiele et al[27]	University of Leipzig-Heart Centre, Leipzig, Germany	August 2000 to December 2003
Korbinian et al[35]	University Hospital Munich, Munich, Germany	November 2015 to March 2018

Question 14: In figure about search strategy please revise numbers, record screened by title and abstract should be 5207 if duplicate are 6670.

Response: Thanks for the helpful suggestion. We are apologized for the lack of clarity in the figure. We edited the Figure 1 as the follow:

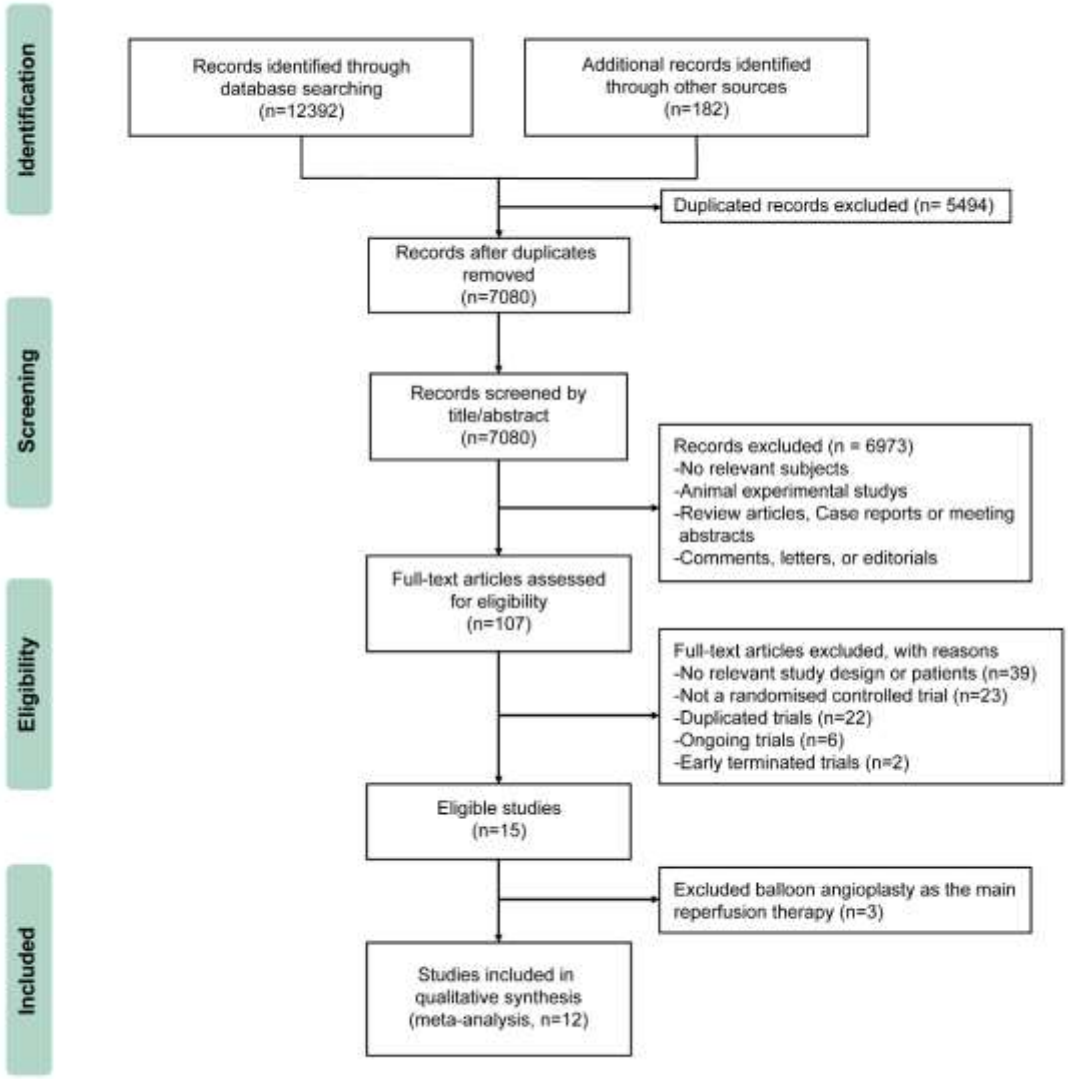


Figure legends:

Figure 1. Flow diagram of study selection. Randomized clinical trials were included.

Question 15: Figures (bottle funnel and forest plot) need caption and legend in supplemental online material, I would suggest to perform a word file with figure legend as supplemental online material.

Response: Thanks for the useful suggestion. We have uploaded a supplementary file in PDF format including caption and legend of supplemental Figures, supplemental Figures and tables. The changes are shown as follows

Online supplementary file 1: PRISMA checklist for the meta-analysis

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6 and Online supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7, and Online supplementary file 3

Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9, Table 1, Table 2 and Online supplementary file 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 and Online supplementary file 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9-11, Figure 2, Figure 3, Online supplementary file 6 and Online supplementary file 7
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12, Online supplementary file 9, Online supplementary file 10 and Online supplementary file 11

Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9-11, Figure 3, Online supplementary file 5 and Online supplementary file 8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	16

Online supplementary file 2: Search strategy

Search strategy in Embase

#	Searches	Results
1	'heart infarction'/exp	409027
2	heart:ab,ti AND infarct*:ab,ti	106516
3	myocard*:ab,ti AND infarct*:ab,ti	306544
4	heart:ab,ti AND attack*:ab,ti	17766
5	cardiovascular:ab,ti AND stroke*:ab,ti	60439
6	stemi:ab,ti	29180
7	nstemi:ab,ti OR 'non stemi':ab,ti OR (non:ab,ti AND stemi:ab,ti)	14111
8	'ischemic heart disease'/exp	709611
9	heart:ab,ti AND ischemi*:ab,ti	120006
10	myocard*:ab,ti AND ischemi*:ab,ti	112169
11	'coronary artery disease'/exp	353023

12	coronary:ab,ti AND artery:ab,ti AND disease*:ab,ti	178679
13	coronary:ab,ti AND atheroscleros*:ab,ti	43833
14	acute:ab,ti AND coronary:ab,ti AND syndrom*:ab,ti	62869
15	acs:ab,ti	43936
16	'angina pectoris'/exp	105403
17	angina*:ab,ti	79942
18	angor:ab,ti AND pectoris:ab,ti	86
19	stenocardia*:ab,ti	927
20	myocardial:ab,ti AND preinfarction:ab,ti AND syndrom*:ab,ti	41
21	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20	1020302
22	'heart assist device'/exp	43026
23	heart:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	18671
24	vascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2623
25	cardiovascular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	1949
26	artificial:ab,ti AND ventricle*:ab,ti	1772
27	ventricle:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	2918
28	ventricular:ab,ti AND assist*:ab,ti AND (device*:ab,ti OR pump*:ab,ti)	24992
29	artificial:ab,ti AND heart:ab,ti AND ventricle*:ab,ti	890
30	blood:ab,ti AND pump*:ab,ti	27560
31	cardiac:ab,ti AND resuscitat*:ab,ti	32204
32	counterpulsation:ab,ti AND device*:ab,ti	646
33	mechanical:ab,ti AND circulat*:ab,ti AND support*:ab,ti	9334
34	cardiac:ab,ti AND assist*:ab,ti AND system*:ab,ti	5468
35	assist*:ab,ti AND circulat*:ab,ti	13513
36	lvad:ab,ti	12954
37	pvad:ab,ti	375
38	tandemheart*:ab,ti	332
39	heartmate*:ab,ti	3663

40	impella*:ab,ti	2192
41	intra:ab,ti AND aortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	6288
42	intraaortic:ab,ti AND balloon:ab,ti AND (pump* OR counterpulsation):ab,ti	2414
43	iabp:ab,ti	4778
44	iabc:ab,ti	172
45	'extracorporeal oxygenation'/exp	27601
46	extracorporeal:ab,ti AND membrane:ab,ti AND oxygenation*:ab,ti	17345
47	extracorporeal:ab,ti AND life:ab,ti AND support*:ab,ti	4998
48	ecmo:ab,ti	17782
49	ecls:ab,ti	2325
50	'heart lung machine'/exp	2137
51	heart:ab,ti AND lung:ab,ti AND machine*:ab,ti	1596
52	'extracorporeal circulation'/exp	73515
53	extracorporeal:ab,ti AND circulat*:ab,ti	15079
54	extracorporeal:ab,ti AND cardiopulmonary:ab,ti AND resuscitation*:ab,ti	1987
55	extracorporeal:ab,ti AND CPR:ab,ti	993
56	ecpr:ab,ti	901
57	#22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56	215006
58	#21 AND #57	39754
59	'animals'/exp NOT 'humans'/exp	5572218
60	'animals, laboratory'/exp	716834
61	'animal experiment'/exp	2667375
62	'animal model'/exp	1461488
63	'rodent'/exp	3906464
64	rat:ti OR rats:ti OR mouse:ti OR mice:ti	1627604
65	#59 OR #60 OR #61 OR #62 OR #63 OR #64	6677800
66	#58 NOT #65	35401

67	'randomized controlled trial':pt OR 'randomized controlled trial'/exp OR 'randomized controlled trials as topic'/exp	839617
68	'controlled clinical trial'/de	432791
69	randomi?ed:ab	880556
70	placebo:ab	310582
71	randomly:ab	466369
72	'clinical trials as topic'/de	109908
73	trial:ti	327712
74	#67 OR #68 OR #69 OR #70 OR #71 OR #72 OR #73	1929647
75	#66 AND #74	3852

Search strategy in Cochrane Library

#	Searches	Results
1	[mh "Myocardial Infarction"]	11198
2	(Myocard* Infarct*):ti,ab,kw	31737
3	(Cardiovascular Stroke*):ti,ab,kw	9663
4	(Heart Attack*):ti,ab,kw	3717
5	(heart infarct*):ti,ab,kw	21977
6	STEMI:ti,ab,kw	3415
7	(NSTEMI or Non-STEMI or (Non and STEMI)):ti,ab,kw	1241
8	[mh "Myocardial Ischemia"]	28522
9	(Myocard* Ischemi*):ti,ab,kw	13500
10	(heart ischemi*):ti,ab,kw	17249
11	[mh "Acute Coronary Syndrome"]	2051
12	(Acute Coronary Syndrom*):ti,ab,kw	7589
13	ACS:ti,ab,kw	4150
14	[mh "Coronary Artery Disease"]	6571
15	(Coronary Artery AND Disease*):ti,ab,kw	23566
16	(Coronary Arterioscleros*):ti,ab,kw	503

17	[mh "Angina Pectoris"]	4588
18	Stenocardia*:ti,ab,kw	64
19	(Angor Pectoris):ti,ab,kw	21
20	angina*:ti,ab,kw	14048
21	(Myocardial Preinfarction Syndrom*):ti,ab,kw	7
22	{or #1-#21}	76205
23	[mh "Heart-Assist Devices"]	180
24	(Heart and Assist* (Device* or pump*)):ti,ab,kw	1280
25	(Vascular and Assist* (Device* or pump*)):ti,ab,kw	265
26	(cardiovascular and Assist* and (Device* or pump*)):ti,ab,kw	307
27	(Artificial Ventricle*):ti,ab,kw	661
28	(Ventricle and Assist* and (Device* or pump*)):ti,ab,kw	241
29	(Ventricular and Assist* and (Device* OR pump*)):ti,ab,kw	764
30	(Artificial Heart Ventricle*):ti,ab,kw	636
31	(blood pump*):ti,ab,kw	5249
32	(cardiac resuscitat*):ti,ab,kw	2945
33	(counterpulsation device*):ti,ab,kw	51
34	(mechanical circulat* support*):ti,ab,kw	340
35	(Cardiac Assist* System*):ti,ab,kw	710
36	(Assist* Circulat*):ti,ab,kw	1077
37	LVAD:ti,ab,kw	346
38	pVAD:ti,ab,kw	14
39	TandemHeart*:ti,ab,kw	9
40	HeartMate*:ti,ab,kw	107
41	Impella*:ti,ab,kw	112
42	[mh "Intra-Aortic Balloon Pumping"]	167
43	(Intra and Aortic and Balloon and (Pump* or counterpulsation)):ti,ab,kw	478
44	(Intraaortica and Balloon and (Pump* or counterpulsation)):ti,ab,kw	4
45	IABP:ti,ab,kw	337
46	IABC:ti,ab,kw	15

47	[mh "Extracorporeal Membrane Oxygenation"]	178
48	(Extracorporeal Membrane Oxygenation*):ti,ab,kw	660
49	(Extracorporeal Life Support*):ti,ab,kw	130
50	ECMO:ti,ab,kw	619
51	ECLS:ti,ab,kw	47
52	[mh "Heart-Lung Machine"]	37
53	(Heart Lung Machine*):ti,ab,kw	262
54	[mh "Extracorporeal Circulation"]	4388
55	(Extracorporeal and Circulat*):ti,ab,kw	1392
56	(extracorporeal cardiopulmonary resuscitation*):ti,ab,kw	48
57	(extracorporeal CPR):ti,ab,kw	23
58	ECPR:ti,ab,kw	20
59	{or #23-#58}	17090
60	#22 and #59	3418
61	[mh "animals"] not [mh "humans"]	61
62	rat or rats or mouse or mice:ti,ab,kw	13725
63	#61 or #62	13782
64	#60 not #63	3379
65	randomized:ti,ab,kw	915787
66	placebo:ti,ab,kw	310693
67	randomly:ti,ab,kw	251577
68	clinical trials:ti,ab,kw	176838
69	{or #65-#68}	1131307
70	#64 and #69	2684

Search strategy in Medline

#	Searches	Results
1	Myocardial Infarction/	167860
2	(Myocard$ and infarct$).ab,ti.	208035

3	(heart and infarct$).ab,ti.	65507
4	(Heart and Attack$).ab,ti.	11461
5	(Cardiovascular and Stroke$).ab,ti.	35842
6	STEMI.ab,ti.	11486
7	(NSTEMI or Non-STEMI or (Non and STEMI)).ab,ti.	4639
8	Myocardial Ischemia/	39284
9	(Myocard$ and Ischemi$).ab,ti.	76017
10	(heart and ischemi$).ab,ti.	73007
11	Acute Coronary Syndrome/	16517
12	(Acute and Coronary and Syndrom$).ab,ti.	35635
13	ACS.ab,ti.	23142
14	Coronary Artery Disease/	64571
15	(Coronary and Artery and Disease$).ab,ti.	112543
16	(Coronary and Arterioscleros$).ab,ti.	1916
17	Angina Pectoris/	32616
18	Stenocardia$.ab,ti.	951
19	(Angor and Pectoris).ab,ti.	45
20	Angina$.ab,ti.	55365
21	(Myocardial and Preinfarction and Syndrom$).ab,ti.	23
22	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21	523082
23	Heart Assist Devices/	14753
24	(Heart and Assist$ and (Device$ or pump$)).ab,ti.	9718
25	(Vascular and Assist$ and (Device$ or pump$)).ab,ti.	1324
26	(Cardiovascular and Assist$ and (Device$ or pump$)).ab,ti.	1090
27	(Artificial and Ventricle$).ab,ti.	1183
28	(Ventricle and Assist$ and (Device$ or pump$)).ab,ti.	1635
29	(Ventricular and Assist$ and (Device$ or pump$)).ab,ti.	13854
30	(Artificial and Heart and Ventricle$).ab,ti.	563
31	(blood and pump$).ab,ti.	17426

32	(cardiac and resuscitat$).ab,ti.	20141
33	(counterpulsation and device$).ab,ti.	414
34	(mechanical and circulat$ and support$).ab,ti.	5182
35	(Cardiac and Assist$ and System$).ab,ti.	3192
36	(Assist$ and Circulat$).ab,ti.	8583
37	LVAD.ab,ti.	4852
38	pVAD.ab,ti.	162
39	TandemHeart$.ab,ti.	193
40	HeartMate$.ab,ti.	1480
41	Impella$.ab,ti.	925
42	Intra Aortic Balloon Pumping/	4056
43	(Intra and Aortic and Balloon AND (Pump* or counterpulsation)).ab,ti.	3937
44	(Intraaortic and Balloon and (Pump* or counterpulsation)).ab,ti.	1667
45	IABP.ab,ti.	2199
46	IABC.ab,ti.	120
47	Extracorporeal Membrane Oxygenation/	11447
48	(Extracorporeal and Membrane and Oxygenation$).ab,ti.	11804
49	(Extracorporeal and Life and Support$).ab,ti.	3112
50	ECMO.ab,ti.	8267
51	ECLS.ab,ti.	1293
52	Heart Lung Machine/	2095
53	(Heart and Lung and Machine$).ab,ti.	1060
54	Extracorporeal Circulation/	12973
55	(Extracorporeal and Circulat*).ab,ti.	11010
56	(extracorporeal and cardiopulmonary and resuscitation*).ab,ti.	1242
57	(extracorporeal and CPR).ab,ti.	427
58	ECPR.ab,ti.	380
59	23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58	100032

60	22 and 59	12723
61	animals/ not humans/	4757950
62	exp Animals, Laboratory/	892881
63	exp Animal Experimentation/	9614
64	exp Models, Animal/	584650
65	exp rodentia/	3282798
66	(rat or rats or mouse or mice).ti.	1355142
67	61 or 62 or 63 or 64 or 65 or 66	5725191
68	60 not 67	11457
69	randomized controlled trial.pt. or randomized controlled trial/ or Randomized Controlled Trials as Topic/	658872
70	controlled clinical trial.pt.	94082
71	placebo.ab.	215311
72	randomly.ab.	351488
73	clinical trials as topic.sh.	194813
74	randomi?ed.ab.	609920
75	trial.ti.	234697
76	69 or 70 or 71 or 72 or 73 or 74 or 75	1439354
77	68 and 76	1339

Search strategy in Pubmed

#	Searches	Results
1	"Myocardial Infarction"[Mesh]	178446
2	(Myocard*[Title/Abstract]) AND Infarct*[Title/Abstract]	213501
3	(Cardiovascular[Title/Abstract]) AND Stroke*[Title/Abstract]	37057
4	(Heart[Title/Abstract]) AND Attack*[Title/Abstract]	11736
5	(heart[Title/Abstract]) AND infarct*[Title/Abstract]	69148
6	STEMI[Title/Abstract]	11856

7	(((NSTEMI[Title/Abstract]) OR Non-STEMI[Title/Abstract])) OR ((Non[Title/Abstract]) AND STEMI[Title/Abstract])	4838
8	"Myocardial Ischemia"[Mesh]	437366
9	(Myocard*[Title/Abstract]) AND Ischemi*[Title/Abstract]	77593
10	(heart[Title/Abstract]) AND ischemi*[Title/Abstract]	74762
11	"Coronary Artery Disease"[Mesh]	64579
12	((Coronary[Title/Abstract]) AND Artery[Title/Abstract]) AND Disease*[Title/Abstract]	117973
13	(Coronary[Title/Abstract]) AND Arterioscleros*[Title/Abstract]	2525
14	"Acute Coronary Syndrome"[Mesh]	16526
15	((Acute[Title/Abstract]) AND Coronary[Title/Abstract]) AND Syndrom*[Title/Abstract]	37318
16	ACS[Title/Abstract]	23712
17	"Angina Pectoris"[Mesh]	43697
18	Stenocardia*[Title/Abstract]	952
19	(Angor[Title/Abstract]) AND Pectoris[Title/Abstract]	58
20	angina*[Title/Abstract]	56696
21	((Myocardial[Title/Abstract]) AND Preinfarction[Title/Abstract]) AND Syndrom*[Title/Abstract]	44
22	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21	617614
23	"Heart-Assist Devices"[Mesh]	14752
24	((Heart[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	10556
25	((Vascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1407
26	((cardiovascular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	1202
27	(Artificial[Title/Abstract]) AND Ventricle*[Title/Abstract]	1255
28	((Ventricle[Title/Abstract]) AND Assist*[Title/Abstract]) AND Device*[Title/Abstract]	1601
29	((Ventricular[Title/Abstract]) AND Assist*[Title/Abstract]) AND ((Device*[Title/Abstract]) OR pump*[Title/Abstract])	14282

30	((Artificial[Title/Abstract]) AND Heart[Title/Abstract]) AND Ventricle*[Title/Abstract]	610
31	(blood[Title/Abstract]) AND pump*[Title/Abstract]	17702
32	(cardiac[Title/Abstract]) AND resuscitat*[Title/Abstract]	21366
33	(counterpulsation[Title/Abstract]) AND device*[Title/Abstract]	451
34	((mechanical[Title/Abstract]) AND circulat*[Title/Abstract]) AND support*[Title/Abstract]	5748
35	((Cardiac[Title/Abstract]) AND Assist*[Title/Abstract]) AND System*[Title/Abstract]	3326
36	((Assist*[Title/Abstract])) AND Circulat*[Title/Abstract]	9223
37	LVAD[Title/Abstract]	5058
38	pVAD[Title/Abstract]	173
39	TandemHeart*[Title/Abstract]	204
40	HeartMate*[Title/Abstract]	1515
41	Impella*[Title/Abstract]	972
42	"Intra-Aortic Balloon Pumping"[Mesh]	4056
43	(((Intra[Title/Abstract]) AND Aortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	4026
44	((Intraaortic[Title/Abstract]) AND Balloon[Title/Abstract]) AND ((Pump*[Title/Abstract]) OR counterpulsation[Title/Abstract])	5487
45	IABP[Title/Abstract]	2253
46	IABC[Title/Abstract]	127
47	"Extracorporeal Membrane Oxygenation"[Mesh]	11452
48	((Extracorporeal[Title/Abstract]) AND Membrane[Title/Abstract]) AND Oxygenation*[Title/Abstract]	12432
49	((Extracorporeal[Title/Abstract]) AND Life[Title/Abstract]) AND Support*[Title/Abstract]	3413
50	ECMO[Title/Abstract]	8706
51	ECLS[Title/Abstract]	1416
52	"Heart-Lung Machine"[Mesh]	2095
53	((Heart[Title/Abstract]) AND Lung[Title/Abstract]) AND Machine*[Title/Abstract]	1081

54	"Extracorporeal Circulation"[Mesh]	84949
55	(Extracorporeal[Title/Abstract]) AND Circulat*[Title/Abstract]	11650
56	((extracorporeal[Title/Abstract]) AND cardiopulmonary[Title/Abstract]) AND resuscitation*[Title/Abstract]	1349
57	(extracorporeal[Title/Abstract]) AND CPR[Title/Abstract]	477
58	ECPR[Title/Abstract]	482
59	#23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56 OR #57 OR #58	158913
60	#22 AND #59	20135
61	((animals[mh]) NOT humans[mh])	4791677
62	Animals, Laboratory[mh]	892913
63	Animal Experimentation[mh]	9615
64	Models, Animal[mh]	584678
65	rodentia[mh]	3282948
66	((rat[Title] OR rats[Title] OR mouse[Title] OR mice[Title]))	1355108
67	#61 OR #62 OR #63 OR #64 OR #65 OR #66	5741699
68	#60 NOT #67	18053
69	((randomized controlled trial[Publication Type]) OR randomized controlled trial[mh]) OR Randomized Controlled Trials as Topic[mh]	661250
70	controlled clinical trial[Publication Type]	614085
71	((randomized[Title/Abstract]) OR randomised[Title/Abstract])	660183
72	placebo[Title/Abstract]	221488
73	randomly[Title/Abstract]	352402
74	trial[Title]	234625
75	#69 OR #70 OR #71 OR #72 OR #73 OR #74	1313992
76	#68 and #75	2105

Search strategy in Web of Science

#	Searches	Results
1	TS=(heart AND infarct*)	345636
2	TS=(myocard* AND infarct*)	452517
3	TS=(heart AND attack*)	44545
4	TS=(cardiovascular AND stroke*)	232512
5	TS=STEMI	20441
6	TS=(NSTEMI OR Non-STEMI OR (Non AND STEMI))	8228
7	TS=(Myocard* AND Ischemi*)	190331
8	TS=(heart AND ischemi*)	276064
9	TS=(Acute AND Coronary AND Syndrom*)	74970
10	TS=ACS	43248
11	TS=(Coronary AND Artery AND Disease*)	351532
12	TS=(Coronary AND Arterioscleros*)	48592
13	TS=Angina*	103224
14	TS=Stenocardia*	1163
15	TS=(Angor AND Pectoris)	472
16	TS=(Myocardial AND Preinfarction AND Syndrom*)	61
17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16	1094596
18	TS=(Heart AND Assist* AND (Device* OR pump*))	34749
19	TS=(Vascular AND Assist* AND (Device* OR pump*))	10298
20	TS=(cardiovascular AND Assist* AND (Device* OR pump*))	21034
21	TS=(Artificial AND Ventricle*)	10067
22	TS=((Ventricular OR Ventricle) AND Assist* AND (Device* OR pump*))	28212
23	TS=(Artificial AND Heart AND Ventricle*)	8861
24	TS=(blood AND pump*)	58329
25	TS=(cardiac AND resuscitat*)	40430
26	TS=(counterpulsation AND device*)	1133
27	TS=(mechanical AND circulat* AND support*)	21978
28	TS=(Cardiac AND Assist* AND System*)	25077

29	TS=(Assist* AND Circulat*)	59084
30	TS=LVAD	9139
31	TS=pVAD	254
32	TS=TandemHeart*	276
33	TS=HeartMate*	2711
34	TS=Impella*	1654
35	TS=(Intra AND Aortic AND Balloon AND (Pump* OR counterpulsation))	7553
36	TS=(Intraaortic AND Balloon AND (Pump* OR counterpulsation))	2998
37	TS=IABP	2862
38	TS=IABC	283
39	TS=(Extracorporeal AND Membrane AND Oxygenation*)	21874
40	TS=(Extracorporeal AND Life AND Support*)	6398
41	TS=ECMO	12728
42	TS=ECLS	2140
43	TS=(Heart AND Lung AND Machine*)	3515
44	TS=(Extracorporeal AND Circulat*)	28102
45	TS=(extracorporeal AND cardiopulmonary AND resuscitation*)	2647
46	TS=(extracorporeal AND CPR)	764
47	TS=ECPR	664
48	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47	250465
49	#17 AND #48	47026
50	TS= clinical trial* OR TS=research design OR TS=comparative stud* OR TS=evaluation stud* OR TS=controlled trial* OR TS=follow-up stud* OR TS=prospective stud* OR TS=random* OR TS=placebo* OR TS=(single blind*) OR TS=(double blind*)	10462135
51	#49 AND #50	16870
52	#49 AND #50 limit clinical trials	2412

Search strategy in ClinicalTrials.gov

Basic search

104 Studies found for: Ventricular Assist Device OR Intra Aortic Balloon Pump OR Extracorporeal Membrane Oxygenation | MI OR Myocardial Infarction OR Coronary Artery Disease

Search strategy in Clinicaltrialsregister.eu

Basic search

78 Studies found for: "ventricular assist device" OR "intra aortic balloon pump" OR "Extracorporeal Membrane Oxygenation"

Online supplementary file 3: The Endpoint Definitions of the Included Studies.

Study	Endpoint	Definition
SEMPER FI[25]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding events according to the Global Use of Strategies to Open Coronary Arteries (GUSTO) trial.
	Arrhythmias	NA.
	Vascular complications	NA.
Zhou et al[29]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	Malignant arrhythmia (not detailed).
	Vascular complications	NA.
IABP-SHOCK II[8,9]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.

	Stroke-TIA	Stroke identified by the presence of new neurologic symptoms in conjunction with signs of ischemia or bleeding on computed tomography.
	Bleeding	Severe or life-threatening bleeding and moderate bleeding as assessed according to the Global Use of Strategies to Open Occluded Coronary Arteries (GUSTO) criteria1.
	Arrhythmias	NA.
	Vascular complications	Peripheral ischemic vascular complications requiring surgical or interventional therapy.
CRISP AMI[28]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	NA.
	Stroke-TIA	Stroke (not detailed).
	Bleeding	Major bleeding events defined by theGlobal Use of Strategies To Open Coronary Arteries (GUSTO) trial and transfusions.
	Arrhythmias	NA.
	Vascular complications	Major vascular complications defined to include limb ischemia requiring intervention, distal embolization, major dissection, pseudoaneurysm or arteriovenousfistula, hematoma larger than 5 cm, and amputations.
Gu et al[30]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Vascular complications (not detailed).
Prondzinsky et al[31]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmia (not detailed).

	Vascular complications	NA.
Arias et al[33]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.
	Vascular complications	NA.
Vijayalakshmi et al[34]	Reinfarction	Reinfarction was defined as chest pain associated with ST-segment re-elevation and an increase in cardiac enzymes (3 times creatine kinase release).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	Bleeding (not detailed).
	Arrhythmias	Arrhythmias including ventricular tachycardia, ventricular fibrillation or atrial fibrillation.
	Vascular complications	Vascular complications (not detailed).
IMPRESS[26]	Reinfarction	Reinfarction Defined by 3rd definition MI. Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following: 1. Symptoms of ischemia. 2. New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB). 3. Development of pathological Q waves in the ECG. 4. Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. 5. Identification of an intracoronary thrombus by angiography or autopsy.
	Repeat revascularization	Repeat PCI or CABG.
	Stroke-TIA	Any stroke confirmed by a neurologist and a concurring CT scan.
	Bleeding	1.Bleeding with associated serum hemoglobin level decrease of at least 5 g/dL (=3.1 mmol/L). The decrease in hemoglobin will be calculated as the last recorded Hb measurement preceding the onset of the bleeding,

		subtracted by the nadir Hb measurement (associated with the bleeding). 2. Bleeding necessitating a minimum of 2 packed cells of blood product transfusion (only blood transfusions that are explicitly related to the bleeding are taken into account). 3. The need for surgery to control the bleeding.
	Arrhythmias	Placement of implantable cardioverter-defibrillator.
	Vascular complications	1.A major bleed arising at the arterial access site (for major bleeding definition see above) requiring extraction of the device. 2. A thrombotic occlusion of the femoral artery. 3. Limb ischemia requiring extraction of either of the study devices. 4. The need for vascular surgery to correct a vascular complication.
Seyfarth et al[32]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	NA.
	Bleeding	Major bleeding (not detailed).
	Arrhythmias	NA.
	Vascular complications	Limb ischemia (not detailed).
Thiele et al[27]	Reinfarction	NA.
	Repeat revascularization	NA.
	Stroke-TIA	Cerebrovascular accidents with neurological dysfunction.
	Bleeding	Major bleedings requiring transfusion of blood components.
	Arrhythmias	NA.
	Vascular complications	Lower extremity ischaemia requiring surgical or interventional action.
Korbinian et al[35]	Reinfarction	Reinfarction (not detailed).
	Repeat revascularization	Repeat revascularization (not detailed).
	Stroke-TIA	NA.
	Bleeding	NA.
	Arrhythmias	NA.

	Vascular complications	NA.

NA indicates not available.

Online supplementary file 4: The Enrollment Information of the Included Studies

Study	Center of Enrollment	Time of Enrollment
SEMPER FI[25]	Catharina Hospital, Eindhoven, Netherlands	August 2014 to December 2018
Zhou et al[29]	Department of Cardiology of Liaocheng People's Hospital, Liaocheng, China	January 2014 to January 2016
IABP-SHOCK II[8,9]	University of Leipzig-Heart Center, Leipzig, Germany Heart Center Bad Krozingen, Bad Krozingen, Germany Asklepios Klinik Langen, Langen, Germany German Heart Center Munich, Munich, Germany Heart Center-Segeberger Kliniken, Bad Segeberg, Germany SLK Kliniken Heilbronn, Heilbronn, Germany University of Greifswald, Greifswald, Germany Martin-Luther-University Halle-Wittenberg, Halle (Saale), Germany Klinikum Links der Weser, Bremen, Germany Heart Center Ludwigshafen, Ludwigshafen, Germany Zentralklinik Bad Berka, Bad Berka, Germany University Clinic of Saarland, Homburg/Saar, Germany Friedrich-Schiller University Jena, Jena, Germany University of Lübeck, Lübeck, Germany Klinik Hennigsdorf, Hennigsdorf, Germany Otto-von-Guericke-University Magdeburg, Magdeburg, Germany Sana Heart Center Cottbus, Cottbus, Germany Heart Center Dresden, Dresden, Germany University of Würzburg, Würzburg, Germany	June 2009 to March 2012

	Hospital München-Neuperlach, Germany	
	Krankenhaus der Barmherzigen Brüder, Regensburg, Germany	
	Charité Berlin Campus Benjamin Franklin, Berlin, Germany	
	Vivantes Klinikum am Urban, Berlin, Germany	
	Carl-Thiem-Klinikum Cottbus, Cottbus, Germany	
	University Rostock, Germany	
	Kerckhoff Klinik Bad Nauheim, Bad Nauheim, Germany	
	Helios Klinikum Erfurt, Erfurt, Germany	
	Charité Berlin-Campus Mitte, Berlin, Germany	
	Ludwigs-Maximilian University Munich-Hospital Großhadern, Germany	
	Vivantes Auguste-Viktoria-Klinikum, Berlin, Germany	
	Klinikum Weiden, Weiden, Germany	
	Klinikum Fulda, Fulda, Germany	
	Katharinenhospital Unna, Unna, Germany	
	Philippusstift Essen, Essen, Germany	
	Carl-von-Basedow-Klinikum Merseburg, Merseburg, Germany	
	Klinikum Lippe-Detmold, Detmold, Germany	
	Schwarzwald-Baar Klinikum VillingenSchwenningen, Villingen-Schwenningen, Germany	
CRISP AMI[28]	Flinders Medical Centre, Bedford Park, Australia Virga Jesse Hospital, Hasselt, Belgium Leeds General Infirmary, Leeds, England Saint Thomas Hospital, London, England Universitatsklinikum Leipzig, Leipzig, Germany University of Freiburg, Freiburg, Germany Klinik III Fur Innere Medizin der Universitat zu Koln, Dresden, Germany CARE Hospital, Hyderabad, India	December 2008 to March 2011

	Baroda Heart Institute and Research Centre, Vadodara, India Sir Ganga Ram Hospital, New Delhi, India Gurunanak CARE Hospital, Hyderabad, India Medanta-The Medicity, Gurgaon, India Tagore Hospital and Heart Care Centre, Jalandhar, India Indraprastha Apollo Hospital, New Delhi, India Apollo Gleneagles Hospital, Kolkata, India Saint John's Medical College and Hospital, Bangalore, India Mater Misercordiae Dublin, Dublin, Ireland Catherina Hospital, Eindhoven, Netherlands Royal Infirmary of Edinburgh, Edinburgh, Scotland Duke University Medical Center, Durham, United States University of Texas Health Science Center, Houston, United States Temple University Hospital, Philadelphia, United States Tufts Medical Center, Boston, United States Newark Beth Israel Medical Center, Newark, United States University of Rochester Medical Center-Strong Memorial Hospital, Rochester, United States Lancaster General Hospital, Lancaster, United States University of Alabama Medical Center, Birmingham, United States Washington Hospital Center, Washington, United States Allegheny General Hospital, Pittsburgh, United States University of Southern California University Hospital, Los Angeles, United States	
Gu et al[30]	Min Hang Hospital, Shanghai, China	January 2005 to February 2009
Prondzinsky et al[31]	Department of Medicine III, Department of Biometrics and Statistics (SU), Halle-Wittenberg, Germany	March 2003 to June 2004

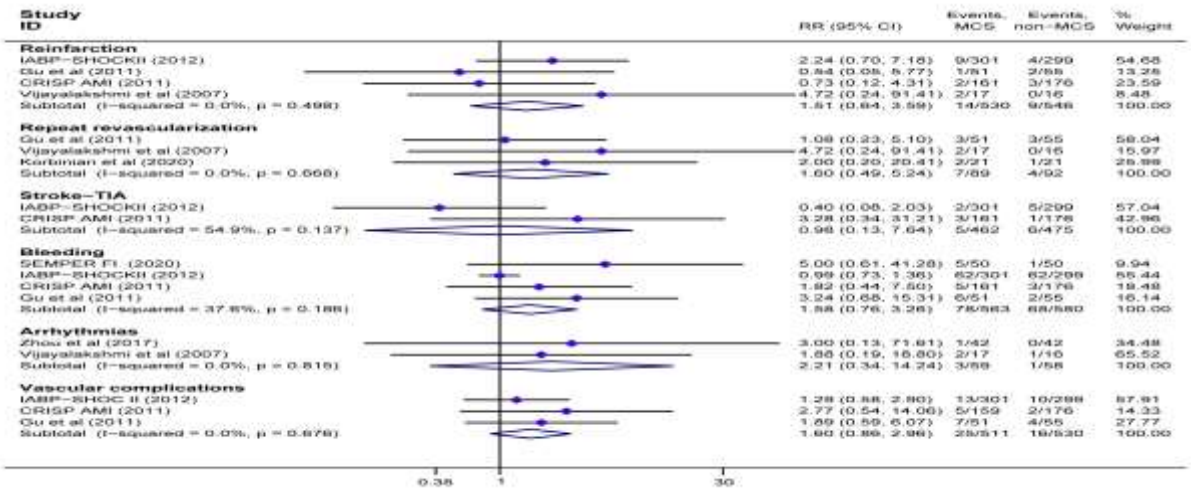
Arias et al[33]	Departamento de Medicina Interna, Fundación Clínica Médica Sur, México	February 2001 to February 2003
Vijayalakshmi et al[34]	The James Cook University Hospital, Middlesbrough, United Kingdom	NA
IMPRESS[26]	Academic Medical Center, Amsterdam, Netherlands Haukeland University Hospital, Bergen, Norway	May 2012 to September 2015
Seyfarth et al[32]	Deutsches Herzzentrum München, Munich, Germany Medizinische Klinik rechts der Isar, Munich, Germany	August 2004 to March 2006
Thiele et al[27]	University of Leipzig-Heart Centre, Leipzig, Germany	August 2000 to December 2003
Korbinian et al[35]	University Hospital Munich, Munich, Germany	November 2015 to March 2018

Online supplementary file 5: Sensitivity analyses of the all-cause mortality by comparison between MCS and non-MCS

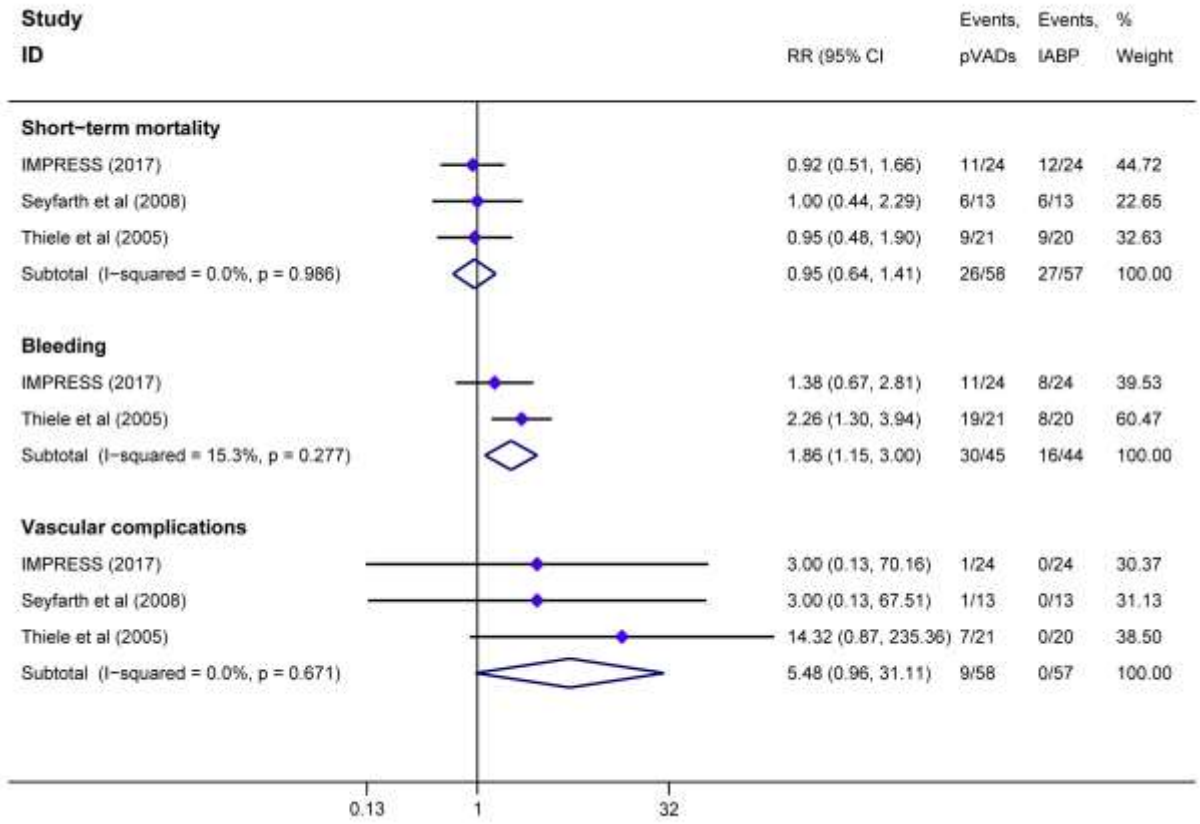
		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	
	In hospital	1.37 (0.50-3.78)
	30 days	0.70 (0.39-1.25)
	Different types of MCS use	
	IABP vs non-IABP	0.92 (0.57-1.47)
	ECLS vs non-ECLS	0.33 (0.01-1.47)
	Excluding high risk bias studies	
	Zhou et al[29], Arias et al[33] and Korbinian et al[35]	0.85 (0.51-1.45)
	Excluding each of the studies	
	SEMPER FI[25]	0.92 (0.57-1.48)
	Zhou et al[29]	0.80 (0.53-1.21)
	IABP-SHOCK II[8]	0.89 (0.45-1.74)
	CRISP AMI[28]	0.95 (0.59-1.55)
	Gu et al[30]	1.02 (0.66-1.58)

	Prondzinsky et al[31]	0.80 (0.48-1.34)
	Arias et al[33]	0.98 (0.58-1.66)
	Vijayalakshmi et al[34]	0.86 (0.55-1.34)
	Korbinian et al[35]	0.92 (0.57-1.47)
Long-term mortality		
	Inconsistent length of follow-up	
	6 months	0.72 (0.43-1.20)
	12 months	0.90 (0.35-2.31)
	Different types of MCS use	
	IABP vs non-IABP	0.87 (0.60-1.25)
	ECLS vs non-ECLS	0.50 (0.18-1.41)
	Excluding high risk bias studies	
	Zhou et al[29] and Korbinian et al[35]	0.72 (0.43-1.20)
	Excluding each of the studies	
	SEMPER FI[25]	0.82 (0.56-1.20)
	Zhou et al[29]	0.70 (0.45-1.09)
	IABP-SHOCK II[9]	0.68 (0.40-1.17)
	CRISP AMI[28]	0.89 (0.65-1.23)
	Gu et al[30]	0.92 (0.66-1.28)
	Korbinian et al[35]	0.87 (0.61-1.25)

CI indicates confidence interval.



Online supplementary file 6: Forest plots for the secondary end points of AMI patients treated with MCS compared with non-MCS. TIA, transient ischaemic attack



Online supplementary file 7: Forest plots for outcomes occurrence comparing pVADs use versus IABP use

Online supplementary file 8: Sensitivity analyses of the all-cause mortality by comparison between pVADs and IABP

		Risk Ratio (95%CI)
Short-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	

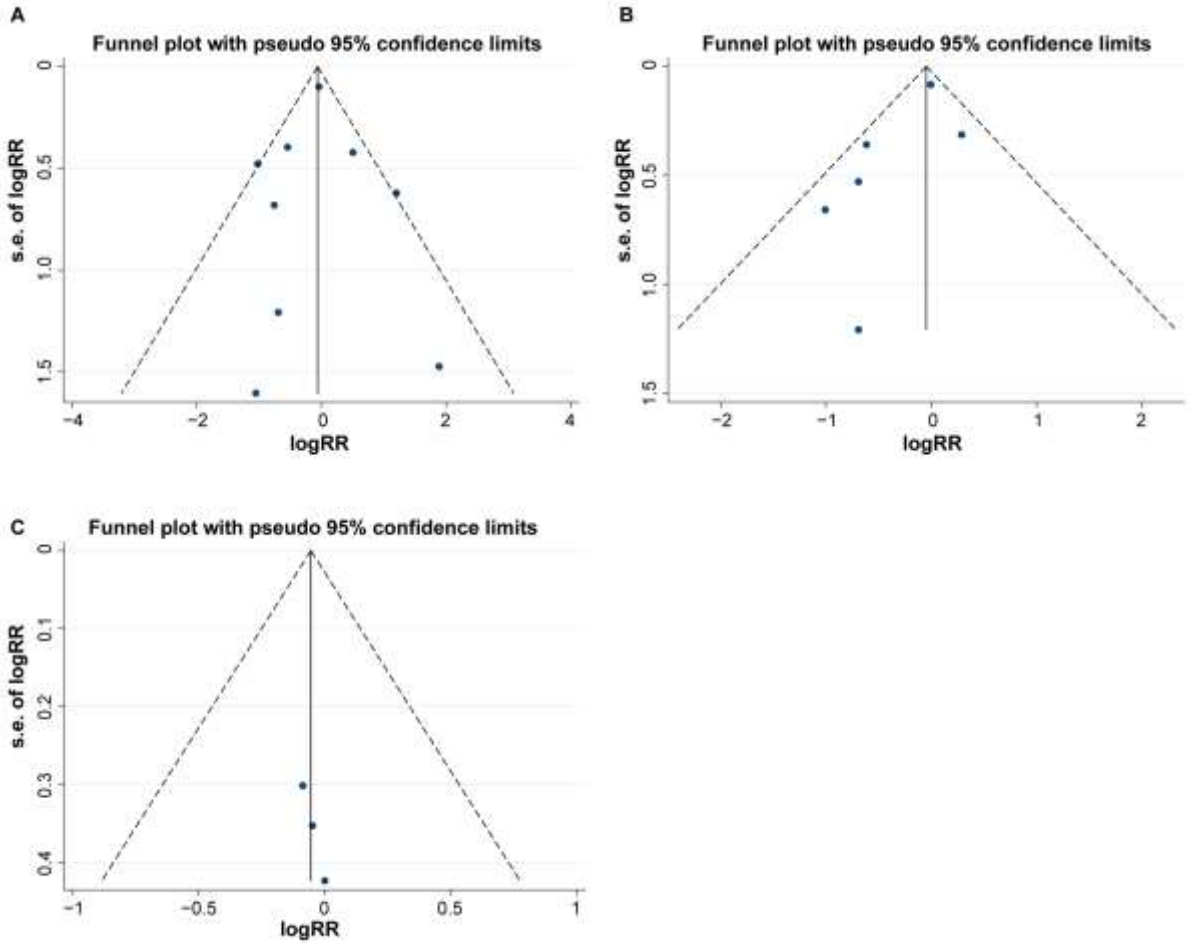
	IMPRESS[26]	0.97 (0.57-1.65)
	Seyfarth et al[32]	0.93 (0.59-1.46)
	Thiele et al[27]	0.94 (0.58-1.53)
Long-term mortality		
	Inconsistent length of follow-up	Not applicable
	Different types of MCS use	Not applicable
	Excluding high risk bias studies	Not applicable
	Excluding each of the studies	Not applicable

Online supplementary file 9: Risk of bias in randomized studies, based on the Cochrane Risk of Bias Tool for Randomized Controlled Trials

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
SEMPER FI[25]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Zhou et al[29]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IABP-SHOCK II[8,9]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
CRISP AMI[28]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Gu et al[30]	Low risk	Unclear risk	Low risk	Low risk	Low risk	Low risk	Low risk
Prondzinsky et al[31]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Arias et al[33]	High risk	High risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Vijayalakshmi et al[34]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
IMPRESS[26]	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Seyfarth et al[32]	Low risk	Unclear risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Thiele et al[27]	Low risk	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk
Korbinian et al[35]	Low risk	Low risk	Low risk	Low risk	Low risk	High risk	Low risk

Low risk
Unclear risk
High risk



Online supplementary file 10: Funnel plot of standard error for: A. MCS compared with non-MCS in short-term mortality; B. MCS compared with non-MCS in long-term mortality; C. pVADs compared with IABP in short-term mortality.

Online supplementary file 11: Publication bias on the association of primary outcome

		Egger's test (P)	Begg's test (P)
MCS vs non-MCS			
	Short-term mortality	0.886	0.917
	Long-term mortality	0.154	0.452
pVAD vs IABP			
	Short-term mortality	0.015*	0.296
	Long-term mortality	Not applicable	Not applicable

* Have statistical difference

Point-by-Point Response to Reviewer 4's Comments

Comments: Authors should be congratulated for their work.

Response: Thanks for the approval of our work. We have tried our best to revise all the mistakes in our manuscript

Reference:

1. Unverzagt S, Buerke M, de Waha A, et al. Intra-aortic balloon pump counterpulsation (IABP) for myocardial infarction complicated by cardiogenic shock. *The Cochrane database of systematic reviews* 2015(3):Cd007398. doi: 10.1002/14651858.CD007398.pub3 [published Online First: 2015/03/31]

2. Ye L, Zheng M, Chen Q, et al. Effects of intra-aortic balloon counterpulsation pump on mortality of acute myocardial infarction. *PloS one* 2014;9(9):e108356. doi: 10.1371/journal.pone.0108356 [published Online First: 2014/10/01]

3. Zheng X-Y, Wang Y, Chen Y, et al. The effectiveness of intra-aortic balloon pump for myocardial infarction in patients with or without cardiogenic shock: a meta-analysis and systematic review. *BMC Cardiovasc Disord* 2016;16(1):148. doi: 10.1186/s12872-016-0323-2

4. Bahekar A, Singh M, Singh S, et al. Cardiovascular outcomes using intra-aortic balloon pump in high-risk acute myocardial infarction with or without cardiogenic shock: a meta-analysis. *Journal of cardiovascular pharmacology and therapeutics* 2012;17(1):44-56. doi: 10.1177/1074248410395019

[published Online First: 2011/02/22]

5. Ahmad Y, Sen S, Shun-Shin MJ, et al. Intra-aortic Balloon Pump Therapy for Acute Myocardial Infarction: A Meta-analysis. *JAMA Intern Med* 2015;175(6):931-39. doi: 10.1001/jamainternmed.2015.0569

6. Smith SC, Jr., Dove JT, Jacobs AK, et al. ACC/AHA guidelines for percutaneous coronary intervention (revision of the 1993 PTCA guidelines)-executive summary: a report of the American College of Cardiology/American Heart Association task force on practice guidelines (Committee to revise the 1993 guidelines for percutaneous transluminal coronary angioplasty) endorsed by the Society for Cardiac Angiography and Interventions. *Circulation* 2001;103(24):3019-41. doi: 10.1161/01.cir.103.24.3019 [published Online First: 2001/06/20]

7. Kaneda H, Hiroe Y, Tanaka S, et al. Long-term outcome of acute myocardial infarction patients treated with stents versus balloon angioplasty: results from randomized trials. *Int J Cardiol* 2009;133(2):241-44. doi: 10.1016/j.ijcard.2007.12.066

VERSION 2 – REVIEW

REVIEWER	Seyfarth, Melchior Univ. Witten-Herdecke, Cardiology
REVIEW RETURNED	26-Mar-2021

GENERAL COMMENTS	The paper was thorougly revised and many inaccuracies deleted.

REVIEWER	Pavasini, Rita Azienda Ospedaliero-Universitaria S. Anna, Cardiovascular Institute
REVIEW RETURNED	23-Mar-2021

GENERAL COMMENTS	The authors well replied to all questions raised. My real concern related to this meta-analysis is which is the novelty compared to the large number of similar studies already present in the literature.

VERSION 2 – AUTHOR RESPONSE

Point-by-Point Response to Reviewer 3's Comments

Comment: The authors well replied to all questions raised. My real concern related to this meta-analysis is which is the novelty compared to the large number of similar studies already present in the literature.

Response: This study highlighted the effect of mechanical circulatory support (MCS) in acute myocardial infarction (AMI) patients **with stenting** revascularization. Pooling the data from twelve studies and 94.5% of AMI patients (1391/1471) with stenting, we are the **first** to identity that MCS

benefits AMI patients with stenting intervention without cardiogenic shock, however, this benefit did not show in the patients complicated with cardiogenic shock.

Previous studies and meta-analyses showed MCS have inconsistent effects in the patients with AMI. The potential reason for this discrepancy may be related to the heterogeneity among included patients. The patients in these studies received various revascularization interventions [1,2,3,4,5,6,7,8,9], including thrombolysis [1,2,4,5,6], balloon angioplasty and stent implantation [3,5,7,8,9], that may lead to different outcomes. Therefore, it is extremely important to further analysis the effect of MCS in AMI patients with distinct revascularization strategy. Since stent implantation of the *infarct*-related coronary artery has been established as a *mainstay* in the *treatment* of *acute myocardial infarction* (*AMI*), this systematic review shed light on the effect of MCS in AMI patients with stenting revascularization.

Changes in the manuscript are listed as below:

The novelty of this meta-analysis has been highlighted in '**Article Summary**' on page 4 and in '**Summary of the evidence**' of '**Discussion**' on page 12 as follows:

This meta-analysis is focusing on the effect of MCS in AMI patients with stenting. (in '**Article Summary**' on page 4)

This meta-analysis is the first to identity that MCS benefits AMI patients with stenting intervention without cardiogenic shock. (in '**Article Summary**' on page 4)

It is the first study that evaluate the efficacy and safety of MCS in patients with AMI undergoing stent implantation. (in '**Summary of the evidence**' of '**Discussion**' on page 12)

Reference:

1. Unverzagt S, Buerke M, de Waha A, et al. Intra-aortic balloon pump counterpulsation (IABP) for myocardial infarction complicated by cardiogenic shock. *The Cochrane database of systematic reviews* 2015(3):Cd007398. 10.1002/14651858.CD007398.pub3

2. Ye L, Zheng M, Chen Q, et al. Effects of intra-aortic balloon counterpulsation pump on mortality of acute myocardial infarction. *PloS one* 2014;9(9):e108356. 10.1371/journal.pone.0108356

3. Zheng XY, Wang Y, Chen Y, et al. The effectiveness of intra-aortic balloon pump for myocardial infarction in patients with or without cardiogenic shock: a meta-analysis and systematic review. *BMC cardiovascular disorders* 2016;16(1):148. 10.1186/s12872-016-0323-2

4. Bahekar A, Singh M, Singh S, et al. Cardiovascular outcomes using intra-aortic balloon pump in high-risk acute myocardial infarction with or without cardiogenic shock: a meta-analysis. *Journal of cardiovascular pharmacology and therapeutics* 2012;17(1):44-56. 10.1177/1074248410395019

5. Ahmad Y, Sen S, Shun-Shin MJ, et al. Intra-aortic Balloon Pump Therapy for Acute Myocardial Infarction: A Meta-analysis. *JAMA internal medicine* 2015;175(6):931-39. 10.1001/jamainternmed.2015.0569

6. Gao ZW, Huang YZ, Zhao HM, et al. Impact of Intra-Aortic Balloon Counterpulsation on Prognosis of Patients with Acute Myocardial Infarction: A Meta-Analysis. *Acta Cardiologica Sinica* 2017;33(6):567-77. 10.6515/acs20170121a

7. Romeo F, Acconcia MC, Sergi D, et al. The outcome of intra-aortic balloon pump support in acute myocardial infarction complicated by cardiogenic shock according to the type of revascularization: a comprehensive meta-analysis. *American heart journal* 2013;165(5):679-92. 10.1016/j.ahj.2013.02.020

8. Su D, Yan B, Guo L, et al. Intra-aortic balloon pump may grant no benefit to improve the mortality of patients with acute myocardial infarction in short and long term: an updated meta-analysis. *Medicine* 2015;94(19):e876. 10.1097/md.0000000000000876

9. Sjauw KD, Engström AE, Vis MM, et al. A systematic review and meta-analysis of intra-aortic balloon pump therapy in ST-elevation myocardial infarction: should we change the guidelines? *European heart journal* 2009;30(4):459-68. 10.1093/eurheartj/ehn602